Expected Returns and Large Language Models *

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Abstract

We extract contextualized representations of news text to predict returns using the state-of-the-art large language models in natural language processing. Unlike the traditional word-based methods, e.g., bag-of-words or word vectors, the contextualized representation captures both the syntax and semantics of text, thus providing a more comprehensive understanding of its meaning. Notably, word-based approaches are more susceptible to errors when negation words are present in news articles. Our study includes data from 16 international equity markets and news articles in 13 different languages, providing polyglot evidence of news-induced return predictability. We observe that information in newswires is incorporated into prices with an inefficient delay that aligns with the limits-to-arbitrage, yet can still be exploited in real-time trading strategies. Additionally, we find that a trading strategy that capitalizes on fresh news alerts results in even higher Sharpe ratios.

Keywords: natural language processing (NLP), large language models, BERT, GPT, OPT, ChatGPT, Bag-of-Words, Word2vec, machine learning, return prediction

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1 Introduction

Economic text is constantly generated by human writers striving to understand and make predictions about economic phenomena. In recent decades, the finance literature has begun to extract information from certain text sources like the financial news press, regulatory filings, and social media. The research agenda of improving economic models through text mining remains in its earliest stages. Research has thus far examined only a limited portion of market-relevant textual data, often focusing on a single specialized data source at a time (e.g., the front page of *The Wall Street Journal*, or the "risk factor" section of 10-K filings). And for each data source, text information is often represented in rudimentary ways (e.g., as a dictionary-based sentiment score or as a "bag of words").

There are good reasons for the limited use of text data to date. Its lack of regular structure makes it far more difficult to work with than standard numeric data sets. Language is an extremely nuanced information encoding scheme. As a result, highly complex models are necessary to faithfully unearth information contained in text. But complex models are prohibitive for many researchers. Technological barriers to entry exclude researchers who lack the specialized skill sets necessary to operate such models. The high computational cost of complex models excludes other researchers who may possess requisite skills but face research funding constraints.

This means that recent textual analysis in finance and economics is the tip of the iceberg. Text is an underexploited data source for understanding asset markets. The challenges of textual analysis today portend an exciting research agenda tomorrow, in which economists gradually expand sourced text corpora and increasingly refine their ability to elicit information from that text.

In this paper we aim to take a step in this direction by constructing refined news text representations derived from large language models (LLMs) and then using these to improve models of expected stock returns. To better understand the role of LLMs, it is helpful to first grasp the current landscape in financial text mining. The most prevalent methods to date are supervised machine learning models that are customized to specific tasks such as forecasting returns (Ke et al. (2019); Jegadeesh and Wu (2013)), volatility (Manela and Moreira (2017)), or macroeconomic conditions (Kelly et al. (2018); Bybee et al. (2020)).

These analyses proceed in two general steps: a text representation step and an econometric modeling step. Step 1 decides on the numerical representation of the text data that will be passed to the Step 2 econometric model. The most common choice in the literature is "bag of words" (BoW), which collapses each document observation into a high dimensional vector of counts spanning all unique terms in the full corpus of documents. In some cases, the numerical representation stops here (e.g., Jegadeesh and Wu (2013); Kelly et al. (2018)). In other cases, the numerical representation is refined further. For example, Ke et al. (2019) reduce the BoW dimensionality from several tens of thousand to a few hundred terms with a correlation screening procedure to filter out irrelevant terms, and Bybee et al. (2020) reduce the dimensionality of counts with an unsupervised topic

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model.[1] The output of Step 1 is a numerical data matrix X of dimension $D \times P$. Rows correspond to the D documents in the text corpus, and each row contains the P-dimensional numerical vector representation of those documents (e.g., P can be the number of terms in a BoW or the number of topics in a topic model). Step 2 uses X as data in an econometric model to describe some economic phenomenon (e.g., return, volatility, and macroeconomic modeling in the references above).

The financial text representations referenced above have some limitations. First, all of these examples begin from a BoW representation, which is overly simplistic and only accesses the information in text that is conveyable by term usage frequency. It sacrifices nearly all information that is conveyed through word ordering or contextual relationships between terms. Second, the ultra-high dimensionality of BoW representations leads to statistical inefficiencies—Step 2 econometric models must include many parameters to process all these terms despite many of the terms conveying negligible information. Dimension reductions like LDA and correlation screening are beneficial because they mitigate the inefficiency of BoW. However, they are derived BoW and thus do not avoid the information loss from relying on term counts in the first place. Third, and more subtly, the dimension-reduced representations are *corpus specific*. For example, when Bybee et al. (2020) build their topic model, the topics are estimated only from *The Wall Street Journal*, despite the fact that topics are general language structures and can be better inferred by using additional text outside of their sample.

Enter the concept of an LLM. LLMs are trained on large text data sets that span many sources and themes. The idea of a LLM is for a specialized research team to perform the Herculean feat of estimating a general purpose language model with astronomical parameterization on truly big text data. LLMs have billions of parameters and are trained on many millions of documents (including huge corpora of complete ebooks, all entries of Wikipedia, and more). But for each LLM, this feat is performed *just once*, then the estimated model is made available for distribution to be deployed by non-specialized researchers in downstream tasks.

In other words, the LLM delegates Step 1 of the procedure above to the handful of people in the world that can best execute it. A Step 2 econometric model can then be built around LLM output. Like LDA (or even BoW), the output of an LLM is a numerical vector representation (or "embedding") of a document. A non-specialized researcher obtains this output by feeding the document of interest through software (which is open-source in many cases). Therefore, an LLM model in Step 1 delivers a numerical matrix X just like the examples above, making it seamless to integrate into Step 2 with little or no modification. The main benefit of an LLM in Step 1 is that it provides more sophisticated and well-trained text representations than used in the literature referenced above. This benefit comes from the expressivity of massive nonlinear model parameterizations and from training on extensive language examples across many domains and from throughout human history. The

[1] Specifically, they employ latent dirichlet allocation (LDA) which can be thought of as a multinomial principal components estimator. This collapses their roughly 20,000-dimensional term count representation for each document to a 180-dimensional topic attention representation.

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transferability of LLMs make this unprecedented scale of knowledge available for finance research.

Our primary research contribution revolves around showcasing the advantages of LLM representations for effectively modeling stock returns. In addition, we compare the performance of LLMs with supervised machine learning models commonly used in the extant finance literature. To achieve this, we undertake two distinct econometric exercises that harness the power of text mining in understanding the financial market. The first exercise involves sentiment analysis, where we extract sentiment information from financial news text and examine how this information is incorporated into the dynamics of stock returns. In the second exercise, we directly leverage the predictive power of financial news text to model the short-term cross-section of expected stock returns.

We study three large-scale pre-trained LLMs: BERT (developed by Google), RoBERTa (by Meta), and OPT (by Meta). We compare this with SESTM, a sentiment analyzer based on BoW representation and trained on task-specific text data (developed by Ke et al. (2019)). We also study a second word-based model, Word2vec (a word-vector representation framework developed by Google). The inputs to our modeling framework are global news text data from Refinitiv in their Thomson Reuters Real-time News Feed (RTRS) and Third Party Archive (3PTY) databases from January 1996 to June 2019. We merge this with individual stock data from CRSP (for US stocks) and Datastream-EIKON (for international stocks).

We find the following main empirical results. First, econometric models that use pre-trained LLM embeddings outperform prevailing text-based machine learning return predictions. This is best summarized in terms of out-of-sample trading strategy performance. A decile spread long-short strategy that buys stocks with high foundation-based return forecasts and sells those with low forecasts earns an annualized Sharpe ratios of 3.5, 4.1, and 4.4 based on BERT, RoBERTa, and OPT models, respectively (gross of trading costs). All three of these significantly outperform the corresponding strategy based on word-embedding forecasts, which earns an annualized Sharpe ratio of 3.2. The BoW strategy by Ke et al. (2019) achieves a Sharpe ratio 3.7, slightly superior to that of BERT, but fail to surpass the results obtained by more advanced LLMs.

Furthermore, we delve into the analysis of the impact of news recency on the relative performance of different models. By focusing on articles labeled as "news alerts" by Refinitiv, we observe that returns remain predictable for significantly longer horizons compared to unflagged articles. Surprisingly, despite the brevity of news alerts, which often consist of only headlines, we find that the distinction between LLMs and word-based models becomes less pronounced. This suggests that the advantages of speed can overshadow differences in language model capacity when it comes to predicting returns based on recent news. In essence, a simple representation of recent news can yield comparable performance to more sophisticated representations of older news. However, as time elapses and the predictive information within the text gradually diminishes, the benefits of employing sophisticated models become comparatively more crucial.

We subsequently demonstrate the relationship between the complexity of LLMs and their performance in predicting returns. By employing a series of OPT models characterized by an escalating

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number of parameters, we illustrate that larger models typically surpass their smaller counterparts in terms of investment performance. The Sharpe ratios yielded by LLMs exhibit greater magnitudes, yet this improvement reaches a saturation point once the number of parameters exceeds 6 billion. This result suggests that while more complex LLMs possess enhanced capabilities in processing text, there is a limit to the benefits gained from increasing their complexity in terms of return prediction.

Our primary findings are derived from the US equity market, focusing on news articles written in English. Additionally, we analyze 16 international stock markets using news articles written in 12 other languages, including Chinese, Japanese, German, Italian, French, Swedish, Danish, Spanish, Finnish, Portuguese, Greek, and Dutch. As a preliminary contribution, we extend the analysis of Ke et al. (2019) to international text. We find similar SESTM performance globally to that documented by Ke et al. (2019) for the US sample. We also find that the BERT and RoBERTa models on average outperforms SESTM. We also note, however, that OPT do not yet provide pre-trained models in multiple languages. This highlights a comparative advantage of simpler to train models like SESTM—they can be easily adapted to different languages without having to wait for enterprise-level training by specialized research teams at leading technology firms.

In a number of extensions, we analyze the differential role of subtle information that is captured by LLMs but missed by BoW models. In particular, we study the role of emotional context of an article and the role of negation words. Both of these tend to be difficult to infer from BoW representations because they rely on a model's ability to extract meaning from ordering and cross-referencing of terms in a document, which is poorly captured with term counts. First, we use LLMs to decompose text into a richer emotional spectrum than extant financial sentiment scores, and show that the performance of "emotion" portfolios aligns with our understanding of human emotion. For example, portfolios representing "approval," "excitement," and "pride" are the best performers while portfolios built on the basis of "fear," "relief," or "surprise" are the worst performers. Then we show that LLMs most strongly outperform BoW models in return prediction when negation terms are more prevalent in a document. The main conclusion from these tests is that LLMs improve over extant financial text mining models because context matters, and LLMs more successfully capture contextual meaning in documents.

The rest of the paper is organized as follows. Section 2 introduces the LLMs and other approaches we compare. Section 3 presents an empirical analysis of stock-level news and return prediction in US and international markets using these methods. Section 4 concludes. The appendix provides additional tables and figures.

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2 The Text Mining Framework

2.1 A Tale of Two Objectives

We employ a supervised approach to mine news text with two primary objectives. The first objective involves sentiment analysis, which entails assessing the tone of a news article. The second objective focuses on predicting the cross-section of returns within a short horizon.

While both sentiment analysis and return prediction illuminate the statistical correlation between news text and returns, they are but components of a broader narrative. We aim to develop trading strategies that can efficiently translate these statistical correlations into profitable investments. The economic impact of these gains serves as a formidable challenge to the efficient market hypothesis.

The efficient market hypothesis, our null hypothesis, posits that expected returns are primarily driven by unpredictable news which is rapidly, and in the most extreme cases instantaneously, assimilated into prices. On the other hand, our research presents an alternative hypothesis: the information contained within news text is not immediately and completely integrated into market prices. This delay might be attributed to factors such as limits-to-arbitrage and rational inattention, suggesting a predictive capacity of news text for future asset price trends.

While the adoption of this alternative hypothesis is largely uncontroversial, its profound significance cannot be overstated. Our predictive analysis provides novel insights into the velocity and magnitude of deviations from the efficient market hypothesis, furnishing fresh evidence to the pool of empirical studies examining this alternative hypothesis.

Sentiment analysis is commonly treated as a classification problem in machine learning. The primary aim is to delineate the relationship between specific text-based features, denoted as $x_{i,t}$, and their associated sentiment labels such as positive or negative, denoted by a binary variable $y_{i,t}$, based on a set of training articles.

The equation below posits the relationship between these labels and features:

$$\mathrm{E}(y_{i,t}|x_{i,t}) = \sigma(x'_{i,t}\beta). \tag{1}$$

In this context, $\sigma(x)$ is a logistic link function, represented as $\sigma(x) = \exp(x)/(1+\exp(x))$. This function has been specifically designed to convert the features into a value ranging from $[0, 1]$, thereby standardizing the quantification of sentiment. This method enables us to derive a sentiment score for any article of the testing sample. The sentiment score quantifies the tone of an article: a score closer to one denotes a stronger positive sentiment.

To accomplish this, we require a sentiment label for each article in the training sample. Each of our news articles is tagged with a corresponding stock and includes a timestamp that records the timing of the news event. Drawing from the methodology presented in Ke et al. (2019), we employ the sign of the stock's return, registered in close temporal relation to a news event, to assign a binary sentiment label (either positive or negative) to the relevant article. Although this label is inherently

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noisy, it is a simple and convenient alternative to manual labeling by expert readers. The empirical analysis conducted by Ke et al. (2019) demonstrates the effectiveness of this approach in measuring news sentiment and its robustness across different return definitions.

Recognizing that news articles often report on events from previous days, we create sentiment labels based on three-day returns, following Ke et al. (2019). This process involves analyzing returns from the day before the article's publication up to the day after. This approach improves the signal-to-noise ratio in sentiment labeling, leading to greater accuracy in the sentiment score — a key goal in sentiment analysis which is to establish a meaningful connection between text and score. It is crucial to note that these three-day returns are utilized solely for in-sample training, thereby avoiding any look-ahead bias when generating sentiment scores for articles within the testing sample.

Sentiment analysis does not directly provide a measurement of expected returns; instead, it merely reflects the tone present in news articles. Hence, we turn to a distinct approach for modeling expected returns, or stated differently, for predicting returns, to examine the extent to which information in news drives the short-term cross-sectional variation of expected returns. The simplest prediction model involves a standard panel regression. The regression equation translates article features, $x_{i,t}$, directly into the corresponding stock's expected return, $\mathrm{E}(r_{i,t+1})$, for the next period:

$$\mathrm{E}(r_{i,t+1}|x_{i,t}) = x'_{i,t}\theta. \tag{2}$$

Inspired by the empirical analysis of Gu et al. (2020), we train this model by collectively considering the next-period returns across all stocks and time periods within our training sample. We can evaluate the effectiveness of our model by assessing its predictive performance in subsequent testing samples.

This pooled panel regression model allows us to represent expected returns as a linear combination of article level features. Similarly, our sentiment model represents the probability of a positive label through a sigmoid function of linear combinations of these features. It is important to note that linearity in model specification is not necessarily the optimal choice. Alternative approaches, such as incorporating neural networks or other nonlinear architectures on top of the $x_{i,t}$s, are certainly viable. However, in order to emphasize the significance of text-based representations and highlight the role they play, we intentionally refrain from introducing such complex nonlinear models. This decision allows us to focus on the simplest model and emphasize the impact of text-based representations.

In the subsequent sections, we elucidate the procedure involved in deriving textual features $x_{i,t}$ from news text through various methodologies, initiating with cutting-edge LLMs in the domain of NLP. This stage is recognized as feature engineering within the parlance of machine learning. Following this, we will present several alternative methods that were proposed prior to the advent of the LLM era.

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2.2 Large Language Models

LLMs represent an innovative approach within the Artificial Intelligence (AI) sphere, first gaining prominence within NLP. This methodology comprises a set of deep learning models, characterized by extensive parameterization and training on expansive datasets.

Distinguishing features of this paradigm pivot around a unique training process devoid of labeled data. Instead, it relies on self-supervised learning techniques. This involves randomly masking words within a text and predicting the masked terms, or through unsupervised language modeling, where the model maximizes the probability of predicting the subsequent sentence based on the current one. Once trained, these LLMs exhibit a remarkable capacity for transfer learning, a process by which the "knowledge" acquired from one task is applied to different tasks. This characteristic enhances their versatility and broadens their applicability across diverse domains.

State-of-the-art LLMs have been dominating performance benchmarks across various NLP tasks, primarily due to their expansive scale. They are often pre-trained on enterprise-level platforms by Google, OpenAI, Meta, etc, some of which have made their pre-trained models publicly available. Our work incorporates three distinct LLMs as benchmarks — Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al. (2018)), Robustly Optimized BERT Pre-training Approach (RoBERTa) (Liu et al. (2019)), and Open Pre-trained Transformers (OPT) by Zhang et al. (2022).

BERT holds a historical significance in the annals of LLMs and NLP as it marked a crucial shift in the creation and application of language models. Before BERT, models predominantly relied on unidirectional or superficially bidirectional understanding of text. BERT brought about a revolution with its deeply bidirectional model, allowing a contextual understanding from both preceding and succeeding words for prediction. This change sparked an influx of research and development in NLP, yielding more advanced models like GPT-2, GPT-3, and RoBERTa. As such, we adopt BERT as our initial benchmark model.

On the other hand, RoBERTa, an offshoot of BERT, was developed by Meta AI. The goal was to augment BERT's performance by modifying its training regimen. Although it shares the foundational architecture with BERT, the variations in the pre-training phase lend RoBERTa its distinct identity. These changes led to substantial performance enhancements, allowing RoBERTa to outdo BERT in numerous NLP benchmark tests. Yet, the question remains whether a model's proficiency in NLP tasks unequivocally translates to stronger performance in investment scenarios. This intriguing query is what our empirical analysis will shed light on.

One remarkable instance of an LLM is ChatGPT, a chatbot that swiftly garnered global recognition. ChatGPT was engineered based on the GPT-3.5 architecture. The initial breakthrough of the GPT model was made by Radford et al. (2018), who introduced a computational framework comprising 117 million parameters. This was subsequently enhanced by Radford et al. (2019) with the introduction of GPT-2, a more robust model featuring a staggering 1.5 billion parameters. Following this, GPT-3 was unveiled in Brown et al. (2020), which saw the model grow to more than

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tenfold the size of GPT-2. Unfortunately, advanced GPT models were not released to the public. As an alternative, we turned to OPT, an open-source implementation of the GPT architecture. OPT has demonstrated performance metrics on par with GPT-3, with its largest model incorporating an enormous 175 billion parameters. Owing to hardware limitations, our studies primarily utilize smaller versions of OPT. Nevertheless, our empirical analysis examines how the scale of the model influences its performance capabilities.

We now turn to details of our implementation from tokenization to feature construction for LLMs.

2.2.1 Tokenization

In any LLM framework, the starting point of a contextualized representation is tokenization. The smallest component of an article is known as a token. Tokens can manifest as characters, words, or subwords, each representing different forms of tokenization. Within LLMs, tokens typically take the form of subwords. Word-based tokenization, which partitions text into individual words based on specific delimiters, is most prevalent. LLMs implement similar tokenization algorithms, which effectively divide rare words into smaller, meaningful subwords. This method of subword tokenization helps to alleviate data sparsity, enabling token reuse and subsequently boosting their frequency of occurrence. Furthermore, it allows for the maintenance of a manageable vocabulary size. This is particularly beneficial given the vast array of different words, or surface forms, present in most languages, especially those that are morphologically rich.[2]

Here's an example from a piece of news regarding Apple: *"The Company also admitted that in addition to macroeconomics in the Chinese market, the price cuts to battery replacements a year earlier to fix the Company's prior surreptitious conduct had hurt iPhone sales."* The BERT tokenizer, which utilizes WordPiece encoding, breaks down this sentence into a sequence of ordered tokens, totaling 43: *'the', 'company', 'also', 'admitted', 'that', 'in', 'addition', 'to', 'macro', '##economic', '##s', 'in', 'the', 'chinese', 'market', ',', 'the', 'price', 'cuts', 'to', 'battery', 'replacements', 'a', 'year', 'earlier', 'to', 'fix', 'the', 'company', '", 's', 'prior', 'sur', '##re', '##pt', '##iti', '##ous', 'conduct', 'had', 'hurt', 'iphone', 'sales', '.'*. In particular, the relatively rare word '*macroeconomics*' is broken down into three tokens, and '*surreptitious*' into five.

While RoBERTa employs the same architectural framework as BERT, it opts for byte-level Byte-Pair Encoding (BPE) for tokenization, akin to GPT-2 as presented by (Radford et al. (2019)). The use of byte-level tokenization enables RoBERTa to more effectively manage out-of-vocabulary words. Notably, OPT also adopts the same BPE tokenizer. Regarding the above example, the BPE tokenizer yields a total of 41 tokens, including punctuations:[3] *'The', 'ĠCompany', 'Ġalso', 'Ġadmitted', 'Ġthat', 'Ġin', 'Ġaddition', 'Ġto', 'Ġmacro', 'econom', 'ics', 'Ġin', 'Ġthe', 'ĠChinese', 'Ġmarket', ',', 'Ġthe', 'Ġprice', 'Ġcuts', 'Ġto', 'Ġbattery', 'Ġreplacements', 'Ġa', 'Ġyear', 'Ġearlier', 'Ġto',*

[2]BERT has 30K tokens, RoBERTa and OPT use about 50K. In contrast, some delimiter based tokenization may result in a vocabulary size over 250K.

[3]A character 'Ġ' is automatically added to represent the space before a word in the original input sentence.

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'Ġfix', 'Ġthe', 'ĠCompany', ''s", 'Ġprior', 'Ġsur', 're', 'pt', 'itious', 'Ġconduct', 'Ġhad', 'Ġhurt', 'ĠiPhone', 'Ġsales', '.'. Similarly, '*macroeconomics*' and '*surreptitious*' are split into multiple tokens.

Both WordPiece and Byte-Pair encoding methods can handle words that are not in their initial vocabulary by breaking them down into smaller, known pieces. In practice, the differences in performance between these two methods tend to be relatively small.

2.2.2 Transformer Architecture

The fundamental architecture of LLMs is rooted in a novel encoder design of deep neural networks, known as the transformer, which was introduced by Vaswani et al. (2017). The transformer encoder maps tokens into vector form, utilizing a series of attention layers, as conceptualized by Bahdanau et al. (2014) and Luong et al. (2015). This enables the modeling of token dependencies, irrespective of their respective positions in the input sequence. By implementing this technique, the traditional recurrent structure, which plays a crucial role in Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) for sequence processing, is effectively eliminated. Notably, the transformer approach bypasses limitations associated with parallelization and memory constraints, hence enhancing the model's scalability.

Although LLMs share core principles, their deep learning architectures differ, each tailored for specific NLP tasks. BERT and RoBERTa employ a bidirectional encoder, which generates contextual representations of tokens by considering both preceding and succeeding instances of their appearances. This architecture proves highly effective for tasks such as sentiment analysis and natural language understanding. On the other hand, OPT solely incorporates a decoder that translates vector embeddings into tokens, making it ideal for applications involving human-like conversation and natural language generation. Consequently, OPT's contextualized embeddings are obtained from a unidirectional architecture.

In our specific context, the distinction between encoder and decoder networks has minimal significance. Both networks possess the capability to generate contextualized embeddings for each token in an input article. These embeddings consider not only the individual tokens themselves but also their respective positions within the article. These contextualized embeddings serve as the fundamental inputs to our procedure for modeling returns. Our empirical analysis reveals that, in terms of the architecture distinction, the impact is relatively minor compared to the complexity of the models, primarily determined by the number of parameters they possess.

2.2.3 Pre-training and Fine-tuning

The training step in this transfer learning context is often termed *pre-training*, serving as a means to an end. This step involves learning about a large number of model parameters (e.g., millions or even trillions) from an extremely large and diverse dataset (e.g., Wikipedia, Common Crawl, WebText). This process allows the model to understand the syntax and semantics of the language, including

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learning the meanings of words, how they are used in different contexts, and the general structure and grammar of the language.

BERT's pre-training process involves two parallel unsupervised tasks: the Masked Language Model (MLM) and Next Sentence Prediction (NSP). In the MLM task, 15% of tokens within the input sequence are randomly hidden, with the model then working to predict these obscured tokens. This MLM goal allows token representation to integrate context from both left and right directions. As implied by its name, NSP attempts to determine if two sentences are consecutive or not.

RoBERTa refines BERT's pre-training by omitting the NSP task and modifying the hyper-parameters, including an extension in training duration, expansion of training data (from 16GB to 160GB of uncompressed texts), an increase in batch-training sizes (from 256 to 8k), and elongation of training sequences. Moreover, while BERT produces static masks just once during data preprocessing, RoBERTa improves upon this by generating masking patterns every time a sequence is input during training. This enhancement supports training across more steps and accommodates larger datasets.

The specific task used for pre-training OPT (and GPT) is called "next token prediction" or "autoregressive language modeling." The model is trained to predict the next word in a sentence given all of the previous words. It does this by learning to understand the context provided by the previous words and using that context to predict the most likely next word. For example, given the input "The Fed raised interest", the model might learn to predict the word "rate" as the next word. The model's parameters are updated during pre-training to minimize the difference between the predicted words and the actual words in the training data.

The adaptation stage, also known as the fine-tuning stage, follows pre-training in the development of LLMs. After the LLM has been pre-trained on a massive corpus of data to understand language in a broad and general sense, it is then adapted or fine-tuned before its deployment to a specific task. This fine-tuning involves training the model on a smaller, task-specific dataset. The tasks can be diverse, such as text classification, sentiment analysis, question answering, summarization, and more. Unlike pre-training, which is unsupervised, fine-tuning is a supervised learning process, as it uses labeled data specific to the task at hand. During fine-tuning, the model's parameters, which were learned during pre-training, are updated to optimize the model's performance on the specific task. This process is usually faster and requires less data than pre-training because the model has already learned a lot of the necessary language understanding during pre-training. In this way, pre-training provides the model with general language understanding, while fine-tuning adapts that understanding to the specific requirements of a task. This two-stage training process has proven to be very effective for building LLMs that perform well across a wide range of NLP tasks.

Drawing inspiration from the work of Peters et al. (2018), we employ a feature extraction approach, also known as probing, by directly utilizing the pre-trained parameters to generate features associated with text data for downstream tasks. To be more specific, we input a new article into the pre-trained model, which results in each token within the article being represented as a vector. These

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vector representations effectively capture the contextual essence of the tokens. These representations are then utilized in subsequent downstream tasks for further exploration and application. While we can perceive the subsequent training stage as a form of fine-tuning, it is important to note that we do not update any parameters produced during the pre-training stage. This streamlined approach minimizes computational efforts, making it easier to replicate our empirical analysis. By adopting this feature extraction approach, we leverage the power of pre-trained models to extract meaningful features from text data without the need for extensive retraining. This efficient process allows us to focus on the downstream tasks at hand while benefiting from the comprehensive contextual understanding encapsulated in the pre-trained parameters.

2.2.4 Article-level Representations

Ultimately, we attempt to construct article level representations, $x_{i,t}$, for subsequent classification and regression tasks. LLMs like BERT and RoBERTa can process input sequences of up to 512 tokens, translating these tokens into a 1024-dimensional vector representation. In contrast, OPT can manage sequences as long as 1,024 tokens and embed each token into a 2,560-dimensional space.[4] In scenarios where an article surpasses 512 (or 1,024) tokens, we simply focus on the initial 512 (or 1,024) tokens. Approximately 60% of US news articles comply with this length restriction. Subsequent empirical analyses suggest that the preliminary 512 tokens effectively encapsulate necessary information for return prediction. After acquiring vector representations for each token, we calculate the vector average across all tokens within an article. The resulting vector is then utilized to represent the entire article's information. Although we employ the mean of all token embeddings to derive an article-level embedding, alternate methods could be considered. For example, it's a common practice to use the embedding of the first token (often referred to as the CLS token) in BERT and RoBERTa, or the last token in OPT, for downstream classification tasks. We have opted for the mean, as it constitutes a reasonable approach for other models like Word2vec, to which we draw comparisons.

2.2.5 Other Fine-Tuned BERT and Multi-Language BERT Models

Several open-source BERT models are available for various tasks. For instance, Araci (2019) fine-tuned a BERT model for a classification task based on the Financial PhraseBank dataset collected by Malo et al. (2014). This dataset includes roughly 5,000 labeled sentences, divided into three categories: positive, neutral, and negative. In a separate work, Yang et al. (2020) pre-trained a different BERT model based on financial communication text. This included Corporate Reports 10-K and 10-Q (comprising 2.5 billion tokens), Earnings Call Transcripts (1.3 billion tokens), and Analyst Reports (1.1 billion tokens), amounting to a total of 4.9 billion tokens in corpus size. This model was later fine-tuned using 10,000 manually annotated sentences (categorized as positive, negative,

[4]Specifically, we chose BERT large, RoBERTa large, and OPT (2.7 billion) as our benchmark set of LLMs. Their total parameters are, respectively, 0.345B, 0.354B, and 2.7B for BERT, RoBERTa, and OPT.

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neutral) from analyst reports. Although this model was pre-trained with data highly relevant to the financial context, it does not leverage the expansive corpus the original BERT was trained on. Due to space constraints, we only provide comparison results based on Yang et al. (2020)'s FinBERT, as our empirical analysis suggests that it surpasses the performance of the model by Araci (2019) (detailed findings not reported).

Beyond English, BERT has been pre-trained with multilingual datasets, enabling its application in the analysis of other languages. Moreover, XLM-RoBERTa, as presented by Conneau et al. (2020) is a multilingual adaptation of RoBERTa. It was pre-trained on 2.5TB of filtered CommonCrawl data encompassing 100 languages. We utilize XLM-RoBERTa large as an extension of RoBERTa in the analysis of non-English languages.

2.3 Word Embeddings

LLMs have evolved beyond an earlier text embedding paradigm that focused on learning morphological word representations as vectors. This progression is rooted in the principles of distributional semantics, as postulated by Harris (1954) and Firth (1957). According to their distributional hypothesis, a word is characterized by the context in which it appears. This idea has been utilized to represent word meanings as vectors, thereby encapsulating semantic similarity in terms of vector similarity. This approach allows for the creation of contextualized embeddings of words in a semantic vector space, capturing the nuanced meaning shifts induced by the contextual environment.

The concept of learning continuous representations of words has a deep-rooted history in NLP, tracing back to the work of Rumelhart et al. (1986). More recently, Mikolov et al. (2013) proposed a simplified approach called Word2Vec that generates high-dimensional vectors on very large corpora. In their work, Mikolov et al. (2013) presented two distinct neural network architectures: the Continuous Bag-Of-Words (CBOW) and the Skip-Gram models. The CBOW model predicts the current word based on its context, excluding the word itself. Conversely, the Skip-Gram model predicts the surrounding words given the current word. An illustrative example by Mikolov et al. (2013) that showcases the efficacy of this approach is the operation vector("King") - vector("Man") + vector("Woman"). Interestingly, the resulting vector from this operation aligns most closely with the vector representation of the word "Queen."

In contrast to LLMs that operate directly on input sequences of variable lengths (up to 512 tokens), Word2Vec employs a fixed-size context window for each word, typically encompassing 5 or 10 words around the current word. This approach limits its capacity to capture contextual information that extends beyond this window. Additionally, Word2Vec is built on a two-layer neural network architecture, a significantly less complex structure compared to the extensive deep neural network architectures employed by foundational models.

For our purposes, we downloaded pre-trained word vectors for English and other languages from

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fastText, an extension of Mikolov et al. (2013)'s Skip-gram model.[5] For English word vectors, we select the model *wiki-news-300d-1M* by Mikolov et al. (2018). This model contains 1 million word vectors trained on Wikipedia 2017, the UMBC webbase corpus, and the statmt.org news dataset, incorporating a total of 16 billion tokens. For non-English languages, the word vectors were trained on Common Crawl and Wikipedia data (Grave et al. (2018)). All these word vectors are 300-dimensional. As we do with LLMs, we calculate the average of all word vectors within a news article to derive the article-level embedding, which is subsequently fed into downstream regressions as features.

2.4 Bag-of-Words

The Bag-of-Words (BOW) model, initially proposed by Harris (1954), represents an article as a vector of word frequencies. This representation takes into account the occurrence and frequency of words, but neglects grammar, word order, and the broader context.

We adhere to the SESTM approach proposed by Ke et al. (2019), which utilizes a structured sentiment model for BOW representations. This method is comprised of three steps. The first step identifies a list of terms (either unigrams or bigrams) most closely correlated with sentiment through a screening process. The second step assigns weights to these words by estimating a topic model. Finally, the third step aggregates these terms into an article-level sentiment score through penalized likelihood estimation. The simplicity, transparency, and theoretical soundness of this approach make it an appropriate BOW benchmark for our purposes.

Among the methods we consider, the SESTM stands out for its simplicity and transparency, but it falls short in accounting for contextual information. On the other hand, LLMs offer the capability to model intricate token connections in natural languages, albeit at the cost of being relatively opaque, often likened to "black boxes." Word2Vec, in comparison, strikes a balance between complexity and capacity, providing context-sensitive embeddings with a simpler architecture. The comparative analysis of these methods offers insights into the degree of predictability derived from a broad spectrum of NLP techniques.

2.5 Why not ChatGPT?

ChatGPT, while a powerful AI tool known for its conversational prowess, is not necessarily ideal for investment purposes. Its primary function as a chatbot indicates that it is not inherently designed for financial forecasting or trading decisions.

For investment purpose, ChatGPT could potentially be deployed as a sentiment analyzer. This application would involve this ChatBot processing news articles to extract sentiment scores, thus interpreting whether the sentiment conveyed in specific news articles is positive or negative. This

[5]FastText, developed by Bojanowski et al. (2017). was chosen due to its multilingual support. Although we adopted the fastText package, we consistently use the term "Word2Vec" for ease of understanding.

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form of sentiment analysis could, in theory, provide valuable insights into public perception of a company, which could help to predict the trajectory of the company's stock returns.

However, using ChatGPT in this way comes with a set of significant challenges. Firstly, the sentiment scores generated by ChatGPT might not be cross-sectionally comparable. When processing news articles, the model evaluates each piece of news on an individual basis, resulting in scores that are relative to that particular article only. This creates a challenge when trying to assess the relative sentiment across a broad range of news articles, as there is no straightforward way to gauge the comparative sentiment magnitudes. Secondly, the practicality of using ChatGPT for real-time trading is questionable. The time required for input/output operations and processing sentiment signals in real time could prove to be too lengthy for fast-paced trading environments. Thirdly, and most significantly, ChatGPT does not offer any measurement of expected returns. This limitation is critical as we attempt to combine LLM's output with additional variables, such as firm fundamentals, technical trends, and macroeconomic market dynamics. These variables, alongside other economic indicators, are critical elements in constructing a comprehensive model for expected returns.

Despite these hurdles, it is worth acknowledging that ChatGPT's impressive, human-like conversational abilities reflect the sophisticated language model's high level of pre-training. If one could gain access to the weight parameters of this language model underlying ChatGPT, i.e., GPT-3.5 or 4.0, it is conceivable that the model could be modified and incorporated for investment purposes, following the same framework proposed here.

3 Empirical Analysis

3.1 Data and pre-processing

3.1.1 Stylized Facts

We have sourced our news text data from Refinitiv, a trusted global provider of financial market data. This dataset encompasses global news from both Thomson Reuters Real-time News Feed (RTRS) and the Third Party Archive (3PTY), spanning from January 1996 to June 2019. For US firms, the news falls into two distinct categories: *articles* and *alerts*. Regular news articles feature both a headline and a body of text, offering a comprehensive narrative of various firm events. In contrast, news alerts focus on delivering timely updates on emerging and unfolding news, and thus consist only of a headline. It is important to note that our US 3PTY database and our international news database do not include alerts. This news text data is then integrated with US equity data from the Center for Research in Security Prices (CRSP), and with international equity data obtained from EIKON (Datastream).

In preparing the news database for analysis, we have implemented several filters. First, we have retained only those news articles and alerts associated with a single stock for which three-day close-to-close returns are available. Furthermore, we have removed excessively short news articles with fewer

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than 100 characters, as well as extremely detailed reports exceeding 100,000 characters. Moreover, we have taken measures to remove redundant articles that essentially replicate the content of preceding stories. Redundancy has been assessed through the computation of a novelty score, derived from cosine similarity calculations based on the bag-of-words representations of any pair of articles. An article is classified as redundant if it attains a cosine similarity score of 0.8 or higher when compared with another article published within the preceding five business days. This process ensures the diversity of the dataset and also safeguards the novelty of the content, resulting in a substantial reduction of superfluous repetition. It is important to note that the removal of such repetition also enhances the signal-to-noise ratio, which is critical as we utilize firm returns as labels in our supervised learning tasks.

Table 1: Summary Statistics of US News Articles and Alerts

Raw Articles			Articles Tagged with Single Stock			Articles With	Filtering Short	Filtering
RTRS	3PTY	Total	RTRS	3PTY	Total	Returns	& Long Articles	Redundancy
6,366,019	4,843,867	11,209,886	2,863,166	4,123,823	6,986,989	4,755,247	4,123,279	3,038,025

Raw Alerts	Alerts Tagged with Single Stock	Alerts With	Filtering	First In	Second In
RTRS	RTRS	Returns	Redundancy	Take Sequence	Take Sequence
4,976,374	4,054,683	3,286,003	2,935,852	1,296,733	522,258

Note: In this table, we report the remaining sample size after each filter applied on the news articles and news alerts on the top and bottom panels, respectively. Columns under "Raw Articles/Alerts" present the numbers of available articles/alerts separately from Thomson Reuters Real-time News Feed (RTRS) and Archive (3PTY). Columns under "Articles/alerts Tagged with Single Stock" presents the number of articles/alerts tagged with a single stock. Columns "Articles/Alerts with Returns" present the number of remaining articles/alerts after matching returns data. Column "Filtering Short & Long Articles" reports the number of articles with at least 100 characters and at most 1,000,000 characters. Columns "First/Second In Take Sequences" report the number of alerts tagged as the first/second in a sequence of developing alerts. Columns "Filtering Redundancy" report the number of remaining articles/alerts after removing those similar to existing ones (cosine similarity score > 0.8) published in the preceding five business days.

Table 1 provides the statistical breakdown of news articles and alerts associated with the US market after applying various filters. The dataset comprises a substantial volume of over 3 million news articles. Notably, a significant proportion of these articles is sourced from third-party news providers. In terms of alerts, the dataset contains approximately 3 million alerts in total. Among these alerts, 55.3% represent the initial news alerts within a sequence of unfolding alerts. 15.9% of the alerts constitute the second in the sequence, while the remaining alerts fall into the category of third or subsequent alerts.

Figure 1 presents the analysis of news articles and alerts' temporal distribution. Both categories share similar patterns, reflecting their intertwined nature. Annual data, displayed in the upper section, reveals a rising trend from 1996 to 2019, with a notable surge in 2008 during the global financial crisis. Monthly patterns, in the middle section, show cyclical peaks in February, May, August, and November, an occurrence likely attributed to concentrated earnings events. It is noticeable that the phenomenon gets especially prominent in alerts. Daily trends, in the lower section, depict increased

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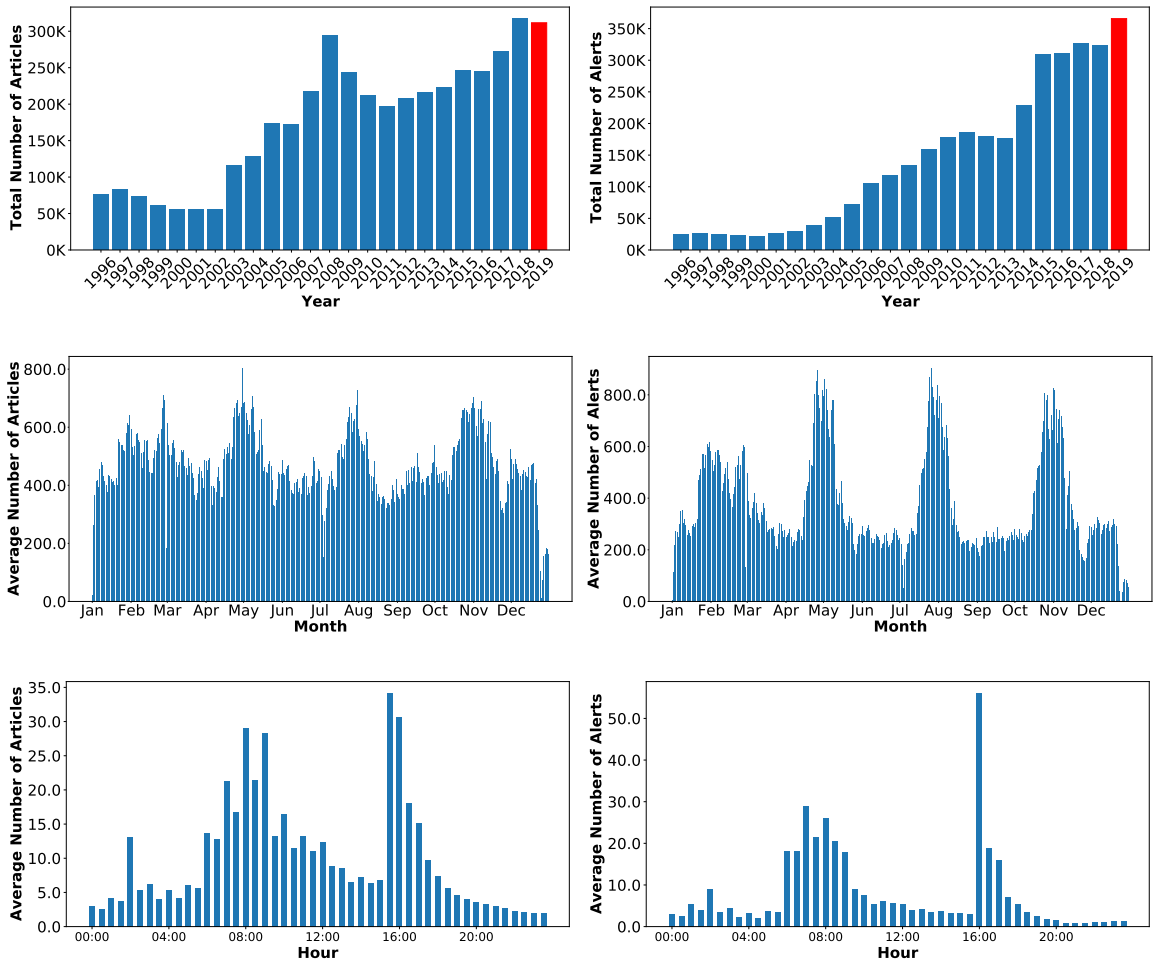
Figure 1: US News Counts



Note: The top panel plots the annual time series of the total number of news articles/alerts, the middle plots the average numbers of news articles/alerts per half an hour (24 hour local time), and the bottom plots the average numbers of news articles/alerts per calendar day. Since our sample ends in June 2019, the number of articles/alerts in 2019 on the first panel is estimated and thus highlighted in red.

news frequency around market opening and closing times, mirroring trading activity ebbs and flows.

Beyond the US market, our analysis also incorporates international markets including China (HK), UK, Australia, Canada, Japan, Germany, Italy, France, Sweden, Denmark, Spain, Finland, Portugal, Greece, and the Netherlands. Figure 2 exhibits annual time series depicting the number of news articles for each international market. A summary of the market information and the processed dataset for each country is encapsulated in Table 2. Table IA1 in the appendix provides a summary of the sample size after undergoing a similar step-by-step filtering process for international markets.

Notably, there exists significant variation in the volume of news articles among different countries, ranging from a minimum of 3,751 articles (Netherlands) to a maximum of 571,285 (UK). While data

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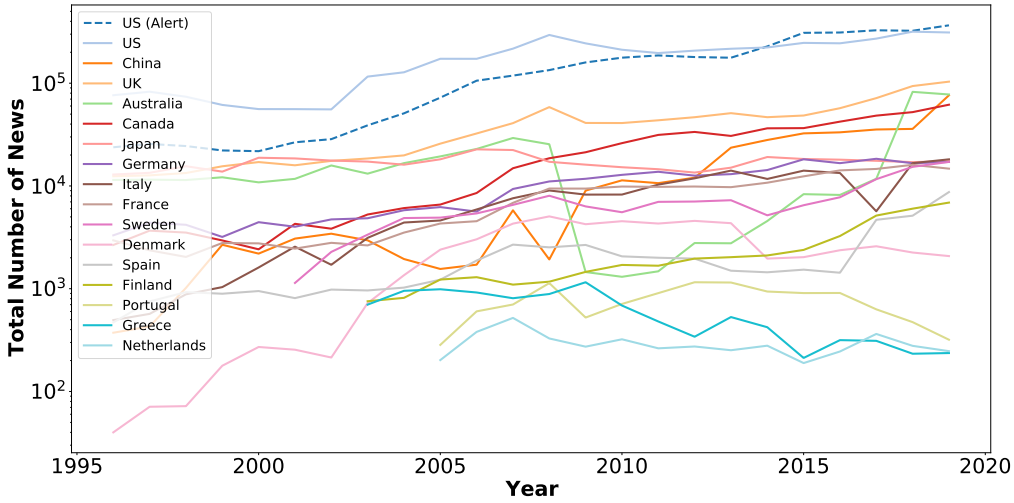
acquisition for most countries commenced in 1996, aligning with the initiation year of US data, certain countries' datasets have later inception points due to gaps in news data provision. For instance, the Netherlands began data collection in September 2005. The monthly average of news-covered stocks varies from a minimum of 11 (Netherlands) to a maximum of 645 (Japan).

Table 2: Summary Statistics of International Markets

	Language	Market Hours	# of Articles	Initial Day	Avg. # of Stocks	# of Days	Avg. # of News
US (Alert)	English	09:30 – 16:00	2,935,852	1996-01-02	1,746	5,929	10,337
US	English	09:30 – 16:00	3,038,025	1996-01-02	2,593	5,933	10,697
UK	English	08:00 – 16:30	571,285	1996-01-02	454	6,087	2,011
Australia	English	10:00 – 16:00	249,190	1996-01-03	287	6,033	877
Canada	English	09:30 – 16:00	350,549	1996-01-03	406	6,032	1,234
China (HK)	Chinese	09:30 – 16:00	182,363	1996-01-03	247	5,768	642
Japan	Japanese	09:00 – 15:00	310,244	1996-01-05	645	5,875	1,092
Germany	German	09:00 – 17:30	178,039	1996-01-03	163	6,031	626
Italy	Italian	09:00 – 17:30	130,168	1996-01-05	97	5,778	458
France	French	09:00 – 17:30	153,779	1996-01-03	167	5,994	541
Sweden	Swedish	09:00 – 17:25	115,195	2001-06-07	170	4,629	526
Denmark	Danish	09:00 – 16:55	43,584	1996-01-22	37	4,559	156
Spain	Spanish	09:00 – 17:30	34,159	1996-01-05	37	5,520	120
Finland	Finnish	10:00 – 18:25	28,633	2003-01-03	50	4,025	143
Portugal	Portuguese	11:30 – 16:30	6,158	2005-05-13	11	2,616	36
Greece	Greek	10:15 – 05:20	7,710	2003-02-19	16	3,057	39
Netherlands	Dutch	09:00 – 17:30	3,751	2005-09-20	11	2,102	22

Note: This table summarizes market information and processed datasets for each country. The columns correspond to the language of news articles, local times corresponding to market hours, the overall count of news articles, the initial day of our sample period, the average number of available stocks per month, the total number of days with news articles, and the average monthly count of news articles.

Figure 2: Total Number of News Articles/Alerts



Note: This figure plots the total number of news articles each year for all international markets. The blue dashed line plots the number for news alerts for US equity market.

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Table 3: Summary Statistics of Characters/Tokens/Words in US News Articles and Alerts

	Article					Alert				
	1%	25%	50%	75%	99%	1%	25%	50%	75%	99%
# of Characters	163	511	1566	3795	29887	32	63	78	103	160
# of OPT Tokens	44	130	349	784	7077	15	26	32	39	60
# of RoBERTa Tokens	43	129	348	783	7076	14	25	31	38	59
# of BERT Tokens	44	129	352	802	7234	9	19	23	28	42
# of Words	6	30	89	240	1896	0	3	5	7	14

Note: Row "# of Characters" report the percentiles of the number of characters in the raw article. Rows "# of OPT Tokens", "# of RoBERTa Tokens", and "# of BERT Tokens" report the percentiles of the number of tokens converted from news text using model specific tokenizer. Row "# of Words" reports the percentiles of the number of words extracted from an article (after removing pronouns/stop words) that are used by SESTM/Word2vec.

Table 4: Summary Statistics of Tokens/Words in International News Articles

	BERT					Word2vec/SESTM				
	1%	25%	50%	75%	99%	1%	25%	50%	75%	99%
US	44	129	352	802	7234	6	30	89	240	1896
UK	56	191	459	950	12424	7	41	111	256	3135
Australia	45	62	165	733	18629	6	18	38	209	4682
Canada	58	271	676	1232	9142	8	60	193	382	2685
China(HK)	32	33	40	73	2983	14	16	21	39	1880
Japan	87	137	212	289	930	71	99	134	159	756
Germany	60	220	359	723	3716	4	50	86	167	954
Italy	44	203	399	1189	7856	8	52	95	199	1707
France	60	221	416	888	7824	12	65	125	272	2163
Sweden	54	161	384	633	3053	18	66	160	263	1349
Denmark	49	187	338	593	2967	10	41	78	144	659
Spain	29	74	187	333	9103	7	18	47	94	1790
Finland	118	296	441	755	9427	44	112	176	315	3258
Portugal	51	185	237	405	1219	4	12	55	113	363
Greece	43	139	222	394	1041	16	58	98	177	468
Netherlands	93	304	523	865	4663	18	84	147	242	1289

Note: Columns under "BERT" report the percentiles of the number of tokens converted from text using specific tokenizers for each country. Columns under "Word2vec/SESTM" report the percentiles of the number of words extracted from an article (after removing pronouns/stop words).

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3.1.2 News Embeddings

We both use word-based models and Large Language Models (LLMs) to generate embeddings for news articles and alerts. LLMs operate at the level of tokens, whereas word-based models take individual words as inputs. Detailed statistics on token and word counts could be found at Table 3, and Table 4 presents the corresponding counts for international news articles.

Word-based models like Word2Vec and SESTM require meticulous data preprocessing to operate effectively at the word level. We follow the procedure outlined in Ke et al. (2019) to derive Bag-of-Words (BOW) representations for news articles. This comprehensive preprocessing includes converting text to lowercase, expanding contractions (e.g., "haven't" to "have n't"), lemmatization (reducing words to their base forms), tokenization, and removal of pronouns, proper nouns, punctuation, special symbols, numbers, non-English words (for English texts), and common stop words like "and," "the," and "is." To operate on text data preprocessing across all languages, we utilize the natural language processing package "spaCy". In this way, each article is then represented using its word count vector, ensuring accurate word-based embeddings.

In contrast, Large Language Models, such as BERT and RoBERTa, possess the advantage of accepting raw, unprocessed text as input. This capability significantly reduces the need for extensive data cleaning. We have selected specific pre-trained LLMs for each country, as detailed in Table IA2 in the Appendix. It is also important to note that for OPT, we only have access to their pre-trained English model, which is available on the Hugging Face model repository.

3.2 Model Training

On the basis of Word2vec and LLMs, we commence by acquiring P-dimensional pre-trained features, denoted as $x_{i,t}$, for each news article i at time t within our sample dataset. In sentiment analysis, we train model (1) using a cross-entropy loss. In With respect to predicting the cross-section of returns, we employ a penalized squared-error loss for training model (2) and additionally apply ridge penalty as a means of regularization for overall robustness of our models.

Notably, SESTM introduces a distinctive structural assumption that sets it apart from conventional word-based models. Consequently, its training and prediction methodologies diverge from the norm. The SESTM framework imposes structural assumptions on the BOW representation of article i at time t, $d_{i,t}$, and its associated sentiment score $p_{i,t}$:

$$\mathrm{P}(sgn(y_{i,t}) = 1) = g(p_{i,t}), \quad d_{[S],i,t} \sim Multinomial(s_{i,t}, p_{i,t}O_+ + (1 - p_{i,t})O_-),$$

where $g(\cdot)$ is some increasing function, $s_{i,t}$ is the total number of sentiment charged words for article i at time t, O_+ and O_- are $|S| \times 1$ vectors of parameters, the set S is part of the vocabulary with an exclusive list of sentiment charged words, and $d_{[S],i,t}$ is a subvector of $d_{i,t}$ with rows corresponding

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to words in set S. Ke et al. (2019) proposes this model and suggests that SESTM's training involves the construction of in-sample estimates for various variables. To be more specific, we can construct $\widehat{S}$ by screening based on how frequently each word appears in a positive article and construct $\widehat{O}_{\pm}$ by running regressions of sentiment word frequencies of each article onto pilot estimates of in-sample sentiment scores. Based on the estimated $\widehat{O}_{\pm}$ and $\widehat{S}$, we are able to conduct the maximum likelihood estimator of the sentiment score for an article out of sample. Several tuning parameters will be involved in the process, including three in the screening step, and one shrinkage parameter in the (penalized) likelihood estimation step.

We train each model using annually updated rolling windows. Each rolling window consists of a 8-year interval for in-sample training with the first six years for training and the next two for validation. The subsequent one-year data is then set aside for out-of-sample testing. As a result, the out-of-sample data range from 2004 to 2019, totaling 16 years. The tuning parameters are selected in the validation sample.

3.3 Sentiment Accuracy with US News

The sentiment analysis aims to predict a binary outcome: one indicating a positive return and zero signifying otherwise. The fitted value of the logistic regression, $\sigma(x'\widehat{\beta})$, is an estimate for the probability of a positive outcome for an article with feature x. A true positive (TP) or true negative (TN) occurs when a predicted "up" probability of greater than 50% coincides with a positive realized return and a probability less than 50% coincides with a negative return.[6] False positives and negatives (FP and FN) are the complementary outcomes. We calculate classification accuracy as follows:

$$\text{Accuracy} = (TP + TN)/(TP + TN + FP + FN).$$

Table 5 presents the yearly out-of-sample prediction accuracy, offering several noteworthy observations. Firstly, the first five models consistently outperform a random guess (50%) in terms of average accuracy over these years, with FinBERT slightly exceeding the random guess. Remarkably, the three Language Models (OPT, RoBERTa, and BERT) exhibit higher overall accuracy compared to the word-based models (Word2vec and SESTM), with OPT achieving an average accuracy of 53.87%.

However, it's important to note that even the best-performing model, OPT, displays relatively low accuracy compared to a random guess. These statistical artifacts are primarily due to market efficiency. In a well-functioning market, unpredictable news dominates equity returns, resulting in a small predictable component. This explains why all models achieve accuracy slightly above 50%. Nevertheless, this modest level of predictability can still lead to substantial gains in terms of

[6]The threshold of 50% is a natural cutoff for positive sentiment score. Alternatively, we also consider the unconditional up probability as a threshold for the data. As we subsequently show, the empirical results remain consistent as we only trade stocks with extreme sentiment scores.

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investment performance.

Table 5: Out-of-Sample Prediction Accuracy

	OPT	RoBERTa	BERT	Word2vec	SESTM	FinBERT
2004	53.91	53.78	53.74	53.50	50.19	49.48
2005	53.26	53.01	53.16	52.62	51.66	50.87
2006	55.24	54.92	54.79	54.73	53.91	50.13
2007	53.87	53.58	53.20	52.49	52.77	50.46
2008	50.56	50.82	50.13	48.42	51.57	53.52
2009	54.39	53.60	53.84	53.80	52.38	49.66
2010	55.23	55.13	55.11	54.97	52.53	48.40
2011	52.79	52.71	51.39	52.36	52.76	50.65
2012	54.24	53.36	53.52	54.07	53.12	49.43
2013	54.71	54.68	54.75	55.61	51.92	47.77
2014	53.86	53.78	53.61	53.47	52.07	49.87
2015	52.89	52.68	52.43	51.50	51.37	51.37
2016	54.10	53.66	53.51	53.06	52.47	49.87
2017	55.57	55.00	54.71	54.34	53.26	49.12
2018	52.05	51.88	51.38	50.76	51.43	51.48
2019	56.66	56.32	56.18	56.44	53.56	48.27
Overall	53.87	53.60	53.38	53.16	52.27	50.08

Note: The table reports out-of-sample classification accuracy of the sentiment measure for OPT, RoBERTa, BERT, Word2vec, SESTM, and FinBERT models for each year in the testing sample.

3.4 Performance of Sentiment Sorted Portfolios

To evaluate out-of-sample predictive performance in economic terms, we introduce a trading strategy that capitalizes on sentiment estimates for investment decisions. Our trading approach is straightforward: we construct a zero-net-investment portfolio by taking long positions in the top quintile (20%) of stocks with the most positive sentiment scores and short positions in the bottom quintile (20%) of stocks with the most negative sentiment scores.

When forming the long and short sides of the strategy, we consider both equal-weighted and value-weighted schemes. Equal weighting provides a straightforward and robust method to assess the predictive power of sentiment across the spectrum of firm sizes and aligns with common practices in hedge funds' news text-based portfolio construction. On the other hand, value weighting places greater emphasis on large-cap stocks, which can be justified for economic reasons (assigning more weight to more productive firms) and practical trade implementation reasons (such as managing transaction costs).

We form portfolios solely at the market open each day, primarily for two reasons. Firstly, acting on overnight news before the morning open is often impractical as it's when most traders gain access to the market. Secondly, unless specialized in high-frequency trading, most funds don't continuously adjust their positions in response to intraday news due to their investment styles and process limitations.

Additionally, in line with a similar choice made by Ke et al. (2019), we exclude articles published between 9:00 am and 9:30 am Eastern Standard Time (EST) for US markets, and a similar strategy

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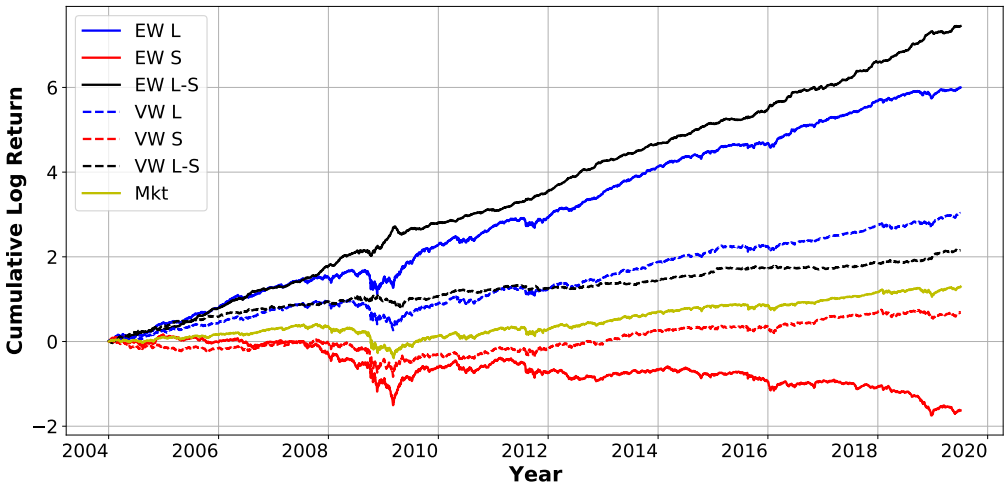
applies to global markets. This decision is driven by our commitment to aligning with realistic considerations. Instead of incorporating the information published during this half-hour window into the same day's prediction input, we opt to shift these articles to the following day. This adjustment allows funds ample time to compute their positions in response to news developments and facilitates trading when market liquidity tends to be at its peak.

Figure 3: One-day-ahead Portfolio Performance based on OPT



Note: This figure compares the out-of-sample cumulative log returns of portfolios sorted on sentiment scores based on the OPT model. The black, blue, and red colors represent the long-short (L-S), long (L), and short (S) portfolios, respectively. The solid and dashed lines represent equal-weighted (EW) and value-weighted (VW) portfolios, respectively. The yellow solid line is the S&P 500 return (SPY).

Figure 3 illustrates the cumulative returns of the long-short (L-S), long (L), and short (S) portfolios constructed using the OPT model. The performance of both equal-weighted ("EW") and value-weighted ("VW") versions of the strategy is also presented for comparison. Notably, the long-short strategy successfully avoids significant drawdowns and demonstrates positive returns during the financial crisis, contrasting with the downward movement of the SPY index. Furthermore, Figure 4 showcases the cumulative one-day trading strategy returns, calculated from open-to-open, based on out-of-sample sentiment forecasts. It is evident that OPT and RoBERTa consistently outperform all word-based models (Word2vec and SESTM). BERT also outperforms both word-based models in the early years until 2012, after which SESTM starts to catch up.

Table 6 provides an overview of the performance statistics for these portfolios. Analyzing these portfolio results in Table 6 reveals three key findings.

Firstly, equal-weighted portfolios exhibit significantly better performance compared to their value-weighted counterparts. Specifically, the long-short strategy based on OPT with equal weights achieves an annualized Sharpe ratio of 4.51, whereas the value-weighted case only attains a Sharpe ratio of 1.24. This indicates that news article sentiment is a stronger predictor of future returns for

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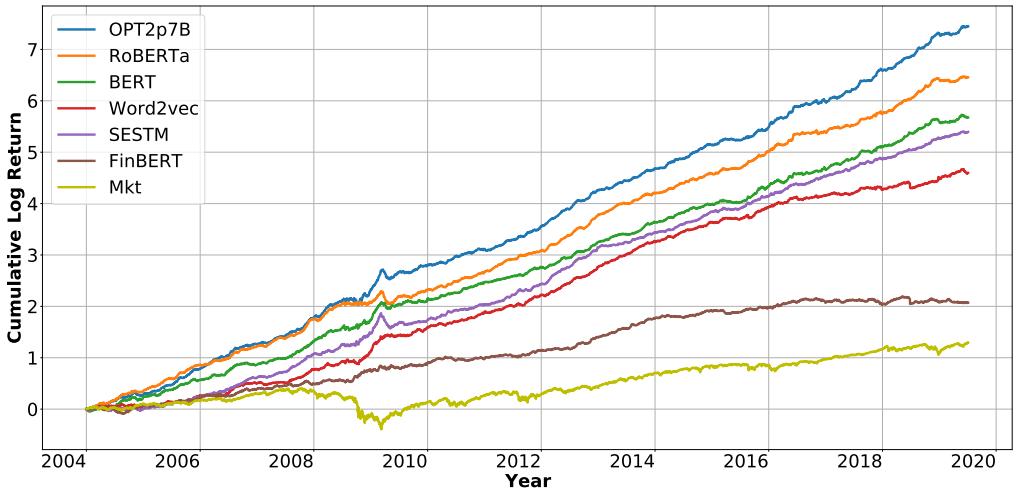
Figure 4: Performance of US Equal-Weighted Portfolios



Note: US equal-weighted long-short portfolio cumulative log returns for portfolios sorted on sentiment scores. The portfolios are built on the basis of OPT, RoBERTa, BERT, Word2vec, SESTM, and FinBERT models, respectively. "Mkt" is the cumulative return for S&P 500 return (SPY).

Table 6: Performance of Daily News Sentiment Portfolios in US

	OPT						RoBERTa					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.41	-0.08	0.49	0.20	0.07	0.14	0.39	-0.03	0.42	0.22	0.09	0.13
Std	0.20	0.23	0.11	0.19	0.22	0.11	0.21	0.22	0.10	0.19	0.21	0.11
SR	2.03	-0.35	4.51	1.04	0.30	1.24	1.90	-0.14	4.10	1.16	0.43	1.19
	BERT						Word2vec					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.37	-0.00	0.37	0.21	0.09	0.12	0.31	0.01	0.30	0.22	0.12	0.10
Std	0.20	0.22	0.10	0.19	0.21	0.11	0.20	0.22	0.10	0.19	0.21	0.11
SR	1.82	-0.00	3.65	1.12	0.42	1.15	1.56	0.06	3.13	1.12	0.58	0.89
	SESTM						FinBERT					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.34	-0.01	0.35	0.21	0.12	0.09	0.35	0.21	0.14	0.20	0.17	0.02
Std	0.20	0.22	0.10	0.19	0.22	0.11	0.21	0.20	0.10	0.18	0.20	0.10
SR	1.69	-0.05	3.61	1.15	0.57	0.81	1.64	1.03	1.44	1.07	0.89	0.22

Note: The table reports the performance of equal-weighted (EW) and value-weighted (VW) long-short (L-S) portfolios sorted on sentiment scores and their long (L) and short (S) legs. The portfolios are built on the basis of OPT, RoBERTa, BERT, Word2vec, SESTM, and FinBERT models, respectively.

small stocks, assuming all other factors remain constant. Several potential economic explanations exist for this observation. It could be attributed to the facts that i) small stocks receive less attention from investors, resulting in slower responses to news; ii) the underlying fundamentals of small stocks are more uncertain and less transparent, requiring more effort to process news and translate it into actionable price assessments; and iii) small stocks are less liquid, necessitating more time for trading

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to incorporate information into prices.

Secondly, both the long and short sides of the trade demonstrate significant profitability. The long side marginally outperforms the short side, exhibiting a Sharpe ratio of 2.03 compared to 0.35 (in the equal-weighted case). This can be partially attributed to the long side naturally capturing the market equity risk premium, while the short side bears the cost of it. Additionally, it is possible that investors face constraints on short sales, preventing negative news from being fully reflected in prices over short time horizons.

Thirdly, it becomes evident that RoBERTa and OPT consistently exhibit superior performance compared to Word2vec, SESTM, and FinBERT. These noteworthy findings highlight the significant impact of the semantic and syntactic information that LLMs capture, ultimately proving beneficial in the investment context. As LLMs continue to advance and grow more sophisticated, it becomes increasingly apparent that the more intricate and robust models in textual analysis tend to yield stronger investment outcomes. For example, while BERT, with its capacity to analyze contextual information, already demonstrates promising results, it is in its ROBERTa version where its true potential is realized. It further enhances the performance of BERT and solidifies its efficacy as a more powerful tool for investment strategies. This suggests that the LLMs' ability to comprehend the contextual nuances of language and capture intricate linguistic patterns holds great value in the realm of investment decisions.

Finally, we provide a battery of results in Table 7 that investigate the sensitivity of our results to varying some of the implementation choices we make in the portfolio formation step. We focus on the performance of the OPT model. To begin with, we limit the number of stocks for trading each day from the top/bottom quintile (20%) to decile (10%). This increases the Sharpe ratios for equal weighted portfolio, from 4.51 to 4.79 (EW). Though the Sharpe ratio for the value-weighted portfolio decreases from 1.30 to 1.12 (VW), the average return increases from 0.15 to 0.17 (VW). Next, we exclude all news that occurs around earning announcement days to examine the effect of non-earning news. Concretely, we select again the top/bottom quintile of stocks on each day t, but avoid those whose earning announcements are scheduled on day $t-1$, day t, and day $t+1$. The resulting Sharpe ratios suggest that a large amount of information in our sentiment scores does not directly stem from earnings reports. For example, OPT's Sharpe ratios are 4.10 (EW) and 1.40 (VW) for OPT, compared with 4.51 (EW) and 1.30 (VW) from the benchmark. Additionally, we experiment with using residual returns rather than raw returns as our supervising labels in the training procedure. Residuals are obtained from time-series regressions of raw returns over a two-year rolling window with respect to either the CAPM model, or the Fama-French three-factor model, or the CAPM model augmented with 17 industry portfolios.[7] The resulting Sharpe ratios are a bit higher than those of the benchmark model: 5.20, 5.33, and 5.66 for equal-weight L-S portfolios, and 1.61, 1.52, 1.66 for value-weight portfolios. This result suggests that the out-of-sample portfolio performance is fairly robust with respect to labels used in the training sample. Overall, our portfolio formation strategy

[7]These 17 industry portfolios (equal-weighted) are obtained from Kenneth French's website.

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Table 7: Performance of Alternative Sentiment Portfolios based on the OPT Model

Type	BenchMark		Trading Top Decile		w/o. Earning Days	
	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.
EW L-S	4.51	0.49	4.79	0.74	4.10	0.43
EW L	2.03	0.41	2.18	0.45	1.93	0.39
EW S	-0.35	-0.08	-1.15	-0.29	-0.21	-0.05
VW L-S	1.30	0.15	1.12	0.17	1.40	0.15
VW L	1.13	0.21	1.13	0.22	1.17	0.22
VW S	0.31	0.07	0.20	0.05	0.31	0.07
Type	CAPM Resid.		FF3 Resid.		CAPM+ind. Resid.	
	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.
EW L-S	5.20	0.54	5.33	0.56	5.66	0.58
EW L	2.22	0.45	2.27	0.46	2.35	0.48
EW S	-0.42	-0.09	-0.43	-0.10	-0.45	-0.10
VW L-S	1.61	0.19	1.52	0.18	1.66	0.19
VW L	1.33	0.24	1.30	0.24	1.42	0.27
VW S	0.23	0.05	0.31	0.07	0.40	0.08

Note: The table reports the performance of equal-weighted (EW) and value-weighted (VW) long-short (L-S) portfolios and their long (L) and short (S) legs, based on the OPT model. The performance measures include (annualized) annual Sharpe ratio and annualized expected returns. We have presented results of the benchmark OPT portfolio in Table 6. Additionally, we report results for alternative portfolios constructed similarly: trading up to 10% stocks per day, excluding firms near their earnings announcement days, using residuals from time series regressions of raw returns (with respect to the CAPM model, the Fama-French three-factor model, or the CAPM model augmented with 17 industry portfolios as factors).

delivers robust out-of-sample performance for all of the implementation choices we examine.

3.5 News Assimilation

Timely information about the market is captured by news, and this information is assimilated into prices rapidly. To assess the speed of information assimilation in economic terms, we utilize a database containing sequences of news alerts and their order in a developing story. We train various models using all the alerts and construct separate portfolios using subsets of news alerts to illustrate the impact of their sequences. Our objective is to evaluate the performance of portfolios formed based on different sequences of news alerts. The comparison of Sharpe ratios between portfolios based on news alerts and news articles is presented in Table 8.

Several noteworthy results emerge from our analysis. Firstly, the utilization of news alerts leads to higher Sharpe ratios compared to portfolios based on news articles. This finding may be somewhat surprising, considering that news alerts typically lack body text, with the entire information contained in the news title. The fact that trading based on news alerts delivers higher Sharpe ratios suggests that speed of information dissemination is more critical than the comprehensiveness of information.

Secondly, we observe a visible decay in Sharpe ratios from portfolios based on the first set of news alerts (TS1) to those based on the second set of news alerts (TS2), and subsequently to the remaining

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Table 8: Sharpe Ratios of Portfolios based on News Articles and News Alerts

	EW					VW				
	Article	Alert				Article	Alert			
		All	TS1	TS2	Rest		All	TS1	TS2	Rest
OPT	4.51	5.70	5.88	3.91	3.63	1.24	2.96	3.00	1.22	0.88
RoBERTa	4.10	5.90	5.63	3.38	3.29	1.19	2.71	2.86	1.11	0.92
BERT	3.65	4.95	4.28	3.31	2.71	1.15	2.05	2.31	0.89	1.01
Word2vec	3.13	5.04	4.63	3.12	2.32	0.89	2.30	2.48	1.00	0.85
SESTM	3.61	4.97	4.54	2.66	2.57	0.81	2.40	2.27	0.69	0.99
FinBERT	1.44	1.63	2.03	-0.05	0.53	0.22	1.51	1.30	0.12	0.22

Note: The table reports Sharpe ratios of the long-short portfolios built on the basis of OPT, RoBERTa, BERT, Word2vec, SESTM, and FinBERT models, respectively. The top panels reports the Sharpe ratios of portfolios based on sentiment scores and the bottom panel reports the Sharpe ratios of portfolios based on predicted returns. Column "TS1" refers to portfolios that only rely on take sequence 1 alerts, "TS2" only take sequence 2 alerts, "Rest" the remaining alerts, and "All" all alerts.

sets. This decay indicates that news is quickly incorporated into prices. By the time we receive a second update on a developing story, the market has already reacted, resulting in a reduction of Sharpe ratios by more than half. This rapid assimilation of news underscores the importance of timely decision-making.

In Figure 5, we present the average returns in basis points per day along with shaded 95% confidence intervals and Sharpe ratios of equal-weighted portfolios for all models. Examining the sentiment portfolios, we observe that sentiment information is effectively assimilated into prices by the beginning of Day 3. Notably, there is a considerable decline in returns from Day 1 to Day 2, particularly for news alerts. This observation aligns with our expectations since our sentiment score predominantly captures the sentiment of fresh news that has not yet been fully incorporated into market prices.

The Sharpe ratios of the equal-weighted portfolios offer additional insights into the risk-adjusted performance of the models. A higher Sharpe ratio indicates a superior risk-adjusted return, reflecting the ability of a portfolio to generate excess returns per unit of risk taken. By considering the sentiment portfolios, we can assess the risk-adjusted performance of news-based investment strategies. The declining Sharpe ratios from Day 1 to Day 3 indicate diminishing risk-adjusted returns as news information becomes fully incorporated into prices.

Overall, the findings from Figure 5 underscore the time-sensitive nature of news-driven market dynamics. It emphasizes the need for timely decision-making and efficient trading strategies to capitalize on news-based signals before they lose their potential profitability due to rapid information assimilation by the market.

3.6 The Virtue of Complexity

A pertinent question often arises in the realm of LLMs: does their performance improve in tandem with their complexity? While there's substantial evidence within the field of Natural Language

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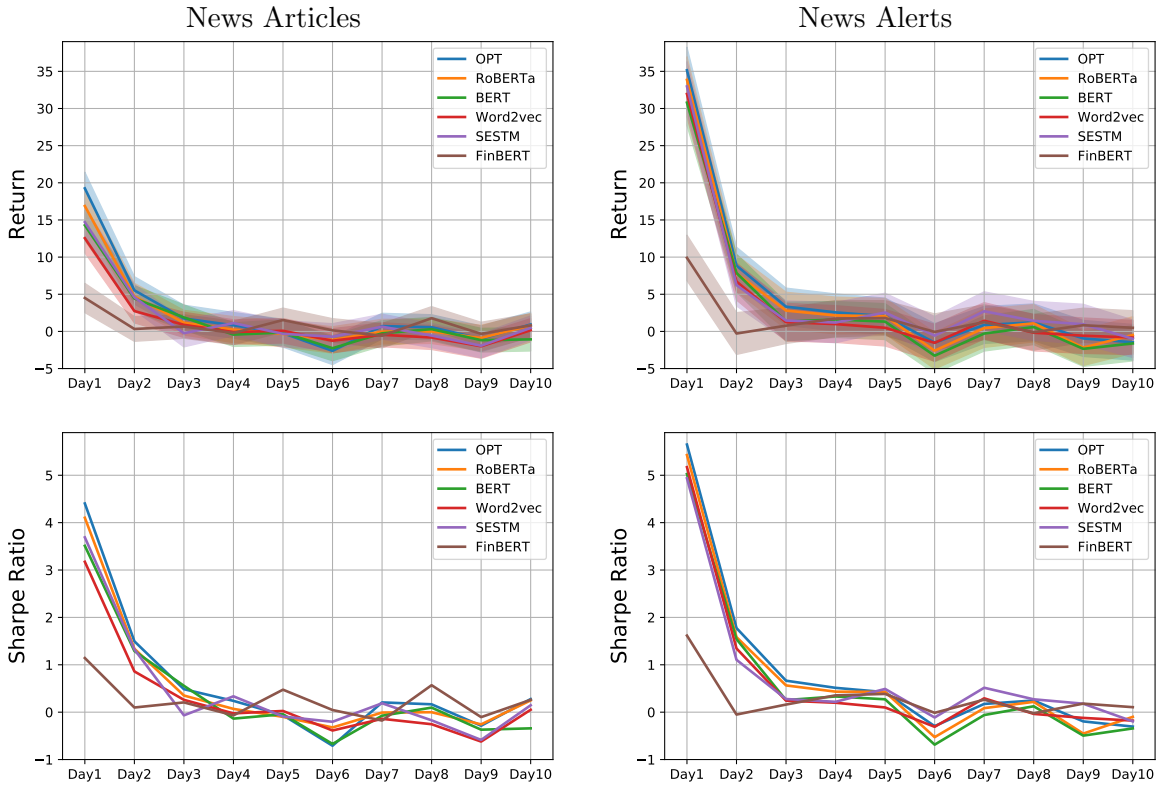
Figure 5: Speed of News Assimilation



Note: This figure compares average one-day holding period returns to the news trading strategy as a function of when the trade is initiated. We consider daily open-to-open returns initiated from one to 10 days following the announcement. We report equal-weighted portfolio average returns (in basis points per day), with 95% confidence intervals given by the shaded regions on the top panel, as well as the corresponding Sharpe ratios on the bottom panel.

Processing (NLP) indicating that more complex LLMs tend to outperform simpler counterparts across various NLP tasks, the performance landscape remains unclear in the downstream tasks. This query is especially relevant in the context of constructing news-based trading strategies, which is a problem characterized by a low signal-to-noise ratio, intricately intertwined with the task of extracting relevant information from news text.

Given the special case in constructing trading strategies, there exists the possibility that simpler models have already effectively captured the predictive signals required for return prediction despite the impressive performance LLMs exhibit in textual analysis. This intriguing scenario suggests that the bottleneck might be more related to return prediction than the textual analysis itself. Consequently, our focus shifts towards evaluating the economic implications associated with the size and complexity of LLMs in the context of return prediction tasks.

To explore the relationship between model complexity and performance, we take advantage of the different sizes available for the OPT models. Due to computational constraints, we include five

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OPT models with parameter counts ranging from 125 million to 6.7 billion.

Table 9 provides a comparison of the portfolio performance of OPT models with varying complexities in the US market. When considering portfolios based on news articles, we observe an increasing trend in performance from models with 125 million to 2.7 billion parameters for sentiment portfolios, with the Sharpe ratio improving from 4.39 to 4.51. Notably, the equal-weighted Sharpe ratio reaches a local maximum with the model containing 2.7 billion parameters, while the subsequent larger model experiences a drop in performance.

In contrast, when examining portfolios based on news alerts, we find that the performance remains relatively consistent across models of all complexities. This finding aligns with the nature of short news snippets, as they do not require overly complex models for effective analysis and prediction. Therefore, the portfolio performance for news alerts is largely unaffected by the intricacies of the model used.

These observations provide valuable insights into the relationship between model complexity and performance for LLMs. While increasing complexity appears to enhance performance for sentiment portfolios based on news articles, it does not yield significant additional benefits for portfolios relying on news alerts. This suggests that the complexity of LLMs may have a more pronounced impact on tasks that require a deeper understanding of longer and more context-rich texts.

Table 9: Sharpe Ratios Comparison between Portfolios Built on Variants of OPT Model

	EW					VW				
	Article	Alert				Article	Alert			
		All	TS1	TS2	Rest		All	TS1	TS2	Rest
Portfolios based on Sentiment Analysis										
OPT (125 Million)	4.39	5.47	5.30	3.32	2.84	0.96	2.82	2.84	1.24	1.19
OPT (350 Million)	4.39	5.25	5.03	2.66	2.62	0.87	2.69	2.71	0.78	1.03
OPT (1.3 Billion)	4.46	5.98	5.85	3.55	3.34	1.09	3.06	3.18	0.77	0.79
OPT (2.7 Billion)	4.51	5.70	5.88	3.91	3.63	1.24	2.96	3.00	1.22	0.88
OPT (6.7 Billion)	4.45	6.11	5.73	3.95	3.49	1.53	2.92	2.99	0.73	0.67

Note: The table reports Sharpe ratios of the quintile long-short portfolios built on OPT models, with the number of parameters 125 million, 350 million, 1.3 billion, 2.7 billion (baseline), and 6.7 billion respectively. The top panels reports the Sharpe ratios of portfolio based on sentiment scores and the bottom panel reports the Sharpe ratios of portfolio based on predicted returns. Column "TS1" refers to portfolios that only rely on take sequence 1 alerts, "TS2" only take sequence 2 alerts, "Rest" the remaining alerts, and "All" all alerts.

3.7 Predicting Returns with Textual Features

While sentiment analysis based on LLMs holds promise for investment, the textual features from news may contain information beyond the sentiment (sign) of returns. In this section, we exploit the ability of these features in measuring the cross-section of expected returns within short horizon[8]. Compared to sentiment analysis, there are two key differences. First, the target variables are directly taken

[8]Specifically, we estimate a ridge regression model using the forward one day's open-to-open returns as dependent variable. The penalty coefficient is tuned from the choices of 1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1e0, 1e1, 5e1, 1e2.

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as realized returns. Second, the magnitude of returns may carry valuable information for selecting textual features. We exclude SESTM and FinBERT in the this section since SESTM is designed to measure contemporary sentiment, and FinBERT is fine-tuned specifically for the sentiment of financial news.

Table 10 reports the out-of-sample prediction correlation on a yearly basis. All four models achieve an positive average correlation persistently throughout these years. Both OPT and RoBERTa have out-of-sample correlation higher than 2%. BERT is slightly behind with overall correlation of 1.77, lower than that of Word2vec at 1.94.

Table 10: Out-of-Sample Cross-Sectional Correlations

	OPT	RoBERTa	BERT	Word2vec
2004	0.52	1.31	1.13	1.75
2005	0.80	1.48	1.24	1.34
2006	0.77	0.35	0.89	0.53
2007	1.53	2.21	1.68	1.48
2008	1.29	1.58	0.78	1.40
2009	1.85	1.80	1.49	2.49
2010	2.17	2.57	2.18	2.39
2011	2.05	1.75	1.39	1.11
2012	2.44	2.27	2.12	2.30
2013	2.31	2.52	2.54	2.08
2014	2.25	2.25	1.80	2.19
2015	2.89	2.61	1.61	2.21
2016	1.88	2.09	1.84	2.22
2017	3.54	3.14	2.58	2.74
2018	3.23	3.03	2.82	2.58
2019	3.33	3.14	2.78	2.61
Overall	2.01	2.10	1.77	1.94

Note: The table reports the time-series average of the cross-sectional rank correlations between the predicted return and the future one day's open-to-open returns for OPT, RoBERTa, BERT, and Word2vec models for each year in the testing sample.

The portfolio construction methodology in this section is similar to that of sentiment portfolios, with one key difference: we use predicted returns as sorting variables instead of sentiment scores. Table 11 presents the performance of portfolios derived from cross-sectional return predictions. The results reveal that all three LLMs outperform their sentiment portfolio counterparts in terms of Sharpe ratios, both in equal-weighted and value-weighted scenarios. (with the exception of value weighted BERT). This suggests that directly incorporating next period realized returns assists in selecting textual features that predict returns.

Specifically, OPT improves from 4.51 (EW) and 1.24 (VW) to 4.84 (EW) and 1.32 (VW), while RoBERTa increases from 4.11 (EW) and 1.19 (VW) to 4.54 (EW) and 1.31 (VW). On the other hand, the word-based model Word2vec experiences a decrease in portfolio performance, declining from 3.18 (EW) and 0.93 (VW) to 2.63 (EW) and 0.98 (VW). This highlights the advantage of LLMs over word-based models in their ability to comprehend contextual information. Since predicting future returns relies on more nuanced information extracted from news content than sentiment analysis alone, word-based models, due to their construction, struggle to capture such subtle predictive signals that LLMs

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excel at.

Table 11: Performance of Portfolios sorted by the Cross-Section of Return Predictions

	OPT						RoBERTa					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.45	-0.05	0.50	0.24	0.10	0.14	0.44	-0.03	0.47	0.25	0.12	0.13
Std	0.20	0.22	0.10	0.20	0.19	0.11	0.21	0.22	0.10	0.20	0.19	0.10
SR	2.22	-0.23	4.84	1.21	0.55	1.32	2.12	-0.16	4.54	1.22	0.63	1.31
	BERT						Word2vec					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.40	-0.00	0.40	0.18	0.15	0.03	0.32	0.03	0.29	0.21	0.14	0.08
Std	0.20	0.22	0.10	0.20	0.19	0.10	0.21	0.22	0.11	0.21	0.18	0.11
SR	1.97	-0.01	4.08	0.91	0.78	0.29	1.55	0.12	2.63	1.02	0.77	0.69

Note: The table reports the performance of equal-weighted (EW) and value-weighted (VW) long-short (L-S) portfolios and their long (L) and short (S) legs. The portfolios are built on the basis of OPT, RoBERTa, BERT, and Word2vec, respectively, using the cross-section of expected returns as sorting variables.

3.8 Polyglot Evidence from International Equity Markets

To further examine the performance of our strategy in international markets, we adopt the strategy in US market and construct zero-net-investment portfolios for each country, wherein we long the top quintile of stocks and short the bottom quintile based on the predicted returns. The Sharpe ratios of these portfolios are presented in Table 12. Remarkably, RoBERTa achieves positive equal-weighted Sharpe ratios for 14 out of the 16 countries analyzed. It outperforms the other three models designed for non-English languages with an average equal-weighted Sharpe ratio of 1.05 (with BERT following at 0.88, Word2vec at 0.68, and SESTM at 0.65).

Even when we exclude the United States, we observe that the average Sharpe ratios of RoBERTa and BERT, the two LLMs, continue to surpass those of the word-based models (Word2vec and SESTM). This indicates that LLMs excel in capturing return predictability across various international markets, outperforming models that rely solely on word-based features.

Furthermore, the state-of-the-art OPT model, which is specifically designed for English-speaking countries, achieves an impressive average equal-weighted Sharpe ratio of 2.13, along with a value-weighted Sharpe ratio of 0.57, outperforming all other models considered in our analysis. This demonstrates the robustness and efficacy of the OPT model in generating superior risk-adjusted returns, in the context of a limited number of English-speaking countries.

The results presented in Table 12 highlight the potential benefits of utilizing LLMs, particularly RoBERTa and OPT, in constructing investment portfolios across international markets. These models exhibit superior performance in terms of risk-adjusted returns, suggesting their efficacy in capturing and exploiting cross-country return predictability. A caveat, though, is that Portugal, Greece, and Netherlands have too few stocks with news, and hence the portfolio performance in

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these countries are less reliable.

Table 12: Sharpe Ratios of Portfolios based on the Cross-Section of Return Predictions

	RoBERTa		BERT		Word2vec		SESTM		OPT		FinBERT	
	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW
US	4.47	1.14	3.93	0.17	2.65	0.68	3.70	0.92	4.59	1.29	1.14	0.20
UK	2.48	0.75	2.39	0.32	2.07	0.74	2.20	0.54	2.38	0.63	1.03	0.47
Australia	0.34	0.07	0.75	-0.09	-0.22	-0.01	0.31	0.50	0.13	0.15	0.03	0.55
Canada	1.73	0.88	1.84	0.76	1.24	0.44	0.90	0.47	1.40	0.21	0.56	0.18
China (HK)	1.35	0.84	0.72	0.12	1.18	0.25	1.11	0.79				
Japan	1.43	0.47	1.39	0.83	0.93	-0.01	-0.84	-0.49				
Germany	0.52	0.23	0.44	0.17	0.71	-0.11	0.53	0.63				
Italy	0.26	0.24	0.32	-0.11	0.29	0.28	-0.16	0.08				
France	1.09	0.15	1.14	0.55	0.56	0.10	0.95	0.54				
Sweden	0.70	0.29	0.76	0.40	0.92	-0.11	-0.12	0.43				
Denmark	0.43	0.33	0.75	-0.13	0.35	-0.27	0.26	0.08				
Spain	-0.38	-0.19	-0.35	-0.14	0.07	-0.67	0.07	0.01				
Finland	0.63	0.74	0.59	0.50	0.69	0.24	1.23	1.26				
Portugal	1.14	1.08	-0.05	-0.23	-1.02	-0.36	0.44	0.52				
Greece	0.63	0.45	-1.05	-1.09	-0.44	-0.38	-0.18	-0.18				
Netherlands	-0.01	-0.28	0.49	-0.48	0.95	-0.36	0.01	0.01				
Mean	1.05	0.45	0.88	0.10	0.68	0.03	0.65	0.38	2.13	0.57	0.69	0.35
Mean (Excluding US)	0.82	0.40	0.68	0.09	0.55	-0.01	0.45	0.34	1.31	0.33	0.54	0.40

Note: The table reports Sharpe ratios of long-short portfolios for international market portfolios. The portfolios are built on the basis of RoBERTa, BERT, Word2vec, SESTM, OPT, and FinBERT models, respectively, using the cross-section of predicted returns as sorting variables.

We further examine the cross-sectional differences observed in the Sharpe ratios. Figure 6 reports the relationship between the average number of stocks with available news per day, plotted on the logarithmic horizontal axis, and the equal-weighted Sharpe ratios of portfolios based on return prediction across all 16 countries, depicted on the vertical axis.

The results deliver several noteworthy insights. Firstly, all four models exhibit positive slopes in this correlation. This implies a consistent positive link between profitability and the number of stocks with news available, irrespective of the specific language model employed.

Zooming in on the United States, which boasts the highest number of stocks available for trade and consequently, the largest pool of stocks with news coverage, we observe the highest Sharpe ratios among all the countries considered. This affirms the notion that a larger pool of stocks accompanied by extensive news coverage provides a fertile ground for generating profitable returns through return prediction strategies. Following the US, the United Kingdom, which has the second-largest number of stocks with news, also demonstrates relatively high Sharpe ratios. This reinforces the idea that larger markets with substantial news-covered stocks are conducive to achieving favorable risk-adjusted returns. In contrast, smaller countries such as the Netherlands and Greece, where the number of stocks with news coverage is less than 10, tend to exhibit negative Sharpe ratios overall. This brings to light the challenges associated with achieving profitability through return prediction strategies in markets characterized by limited stock availability.

In sum, these findings underscore the pivotal role played by stock availability and news coverage in shaping the profitability of return prediction strategies across different countries. Larger markets,

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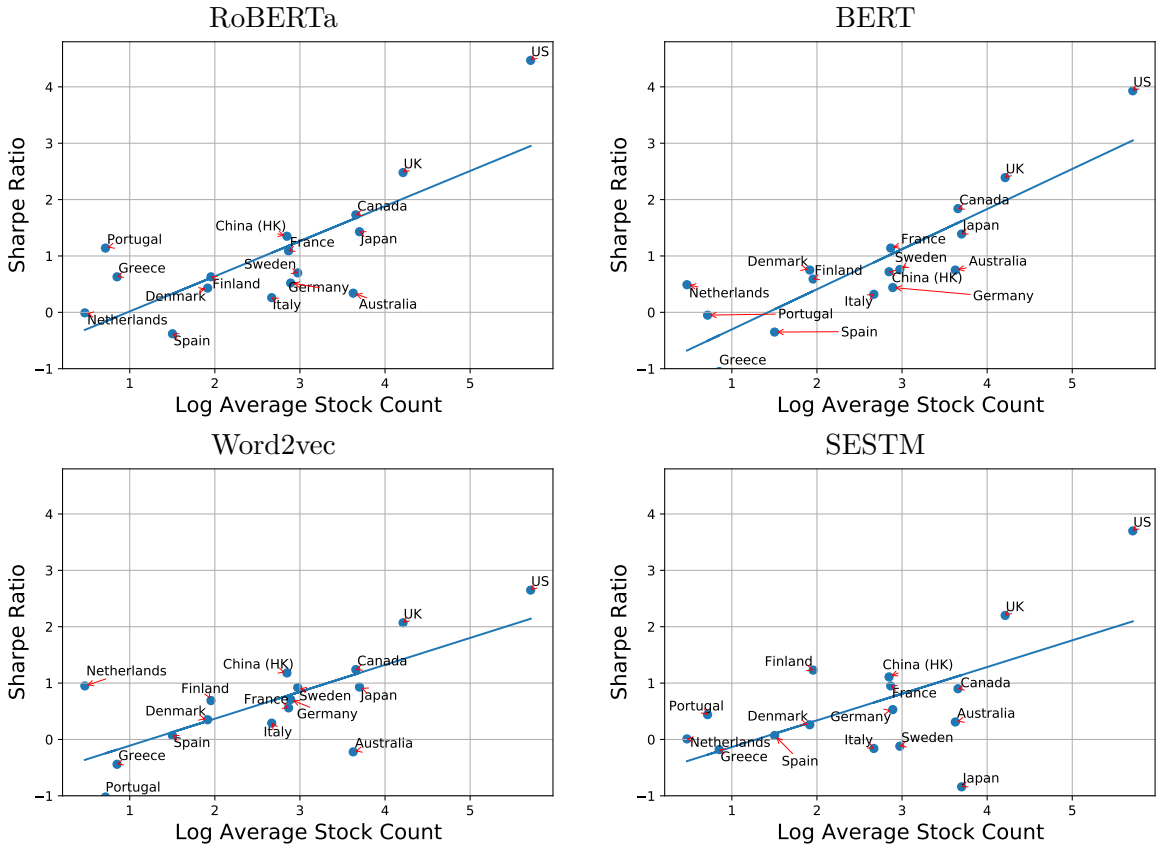
with their multitude of stocks and comprehensive news coverage, provide an advantageous environment for generating positive risk-adjusted returns. Conversely, smaller markets, where stocks are scarce and news coverage is limited, may pose hurdles in the pursuit of profitability through return prediction strategies.

Figure 6: Sharpe Ratios vs the Number of Stocks for International Markets



Note: The figures show annualized out-of-sample Sharpe ratios of equal-weight portfolios versus the logarithm of the average number of stocks available at each rebalance for each country.

3.9 News Emotion Portfolios

To further demonstrate the advantage of LLMs over word-based methods, we utilize the pre-trained BERT model introduced by Demszky et al. (2020) to extract emotion scores from US news articles. These scores are then used to construct long-short portfolios based on their rankings. The BERT model is trained on a dataset called GoEmotions, which consists of approximately 58,000 manually annotated English Reddit comments. The dataset is labeled for 27 distinct emotion categories, including admiration, amusement, anger, annoyance, approval, caring, confusion, curiosity, desire, disappointment, disapproval, disgust, embarrassment, excitement, fear, gratitude, grief, joy, love, nervousness, optimism, pride, realization, relief, remorse, sadness, and surprise. This taxonomy

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builds upon Ekman's six basic emotion categories, encompassing anger, surprise, disgust, enjoyment, fear, and sadness. The taxonomy was further extended through a multi-round iterative process, incorporating feedback from annotators and removing similar or scarce-selected emotions.

To obtain emotion scores for each article, we apply the pre-trained models and calculate the cross-sectional correlation between each emotion score and three-day returns across all stocks, encompassing the entire sample period. This correlation analysis allows us to examine the relationship between specific emotions expressed in news articles and subsequent stock returns. Moreover, we construct next-day long-short portfolios based on each emotion score, which enables us to translate the statistical correlations into economic terms and assess their impact on portfolio performance.

Table 13 reveals intriguing findings. Negative emotions such as disappointment, sadness, embarrassment, remorse, grief, disapproval, and nervousness tend to exhibit negative correlations and Sharpe ratios, while positive emotions such as approval, excitement, joy, gratitude, admiration, amusement, optimism, pride, and love demonstrate the opposite pattern. Among the positive sentiments, eleven (approval, excitement, joy, gratitude, admiration, amusement, optimism, pride, love, desire, and relief) exhibit significant positive t-statistics, indicating their impact on future returns is statistically significant at a 5% significance level. Conversely, seven negative sentiments (disappointment, sadness, embarrassment, remorse, grief, disapproval, and nervousness) yield significant negative t-statistics. The top performer, "approval," achieves an equal-weighted Sharpe ratio of 1.00, while "disappointment" yields an equal-weighted Sharpe ratio of -0.53.

Another noteworthy observation is that although equal-weighted Sharpe ratios generally align with the t-statistics of correlations, the value-weighted Sharpe ratios are predominantly close to zero. This suggests that the emotional content of news predominantly affects the future returns of smaller stocks, as they tend to be more volatile and illiquid.

To provide a deeper understanding of the emotions, we present news examples corresponding to nine specific emotions in Figures 7 and 8. These examples offer a glimpse into the sentiment and tone associated with each emotion category, further illustrating their potential impact on subsequent stock returns.

3.10 Interpreting Textual Analysis

At a short horizon, fundamental signals barely update. Traders often resort to technical signals that rely on price and volume patterns for trading, see, e.g., Jiang et al. (2023). In comparison to technical signals, news provides a more interpretable and transparent source of information. Recognizing the significance of this aspect, we dedicate this section to discussing several methods employed to validate and enhance our understanding of the empirical results obtained from utilizing news data.

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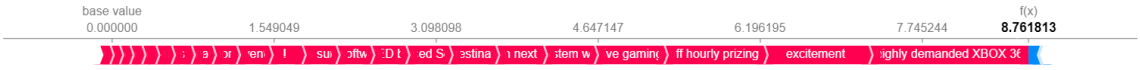
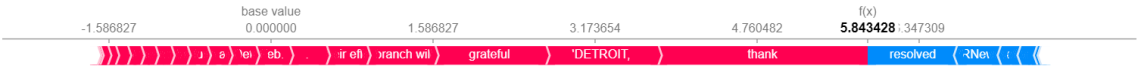
Figure 7: Examples of Positive Emotions in News

(a) Approval

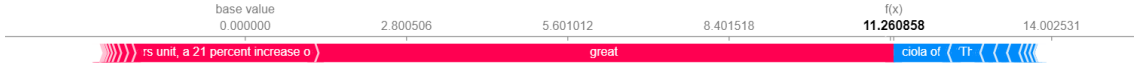
(b) Excitement



(c) Gratitude



(d) Admiration



(e) Joy

Note: This table lists five positive emotion examples from the pre-trained BERT model from Demszky et al. (2020). Segments that contribute to positive SHAP values are highlighted in red and segments with negative SHAP values are highlighted in blue.

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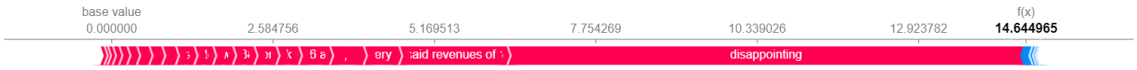
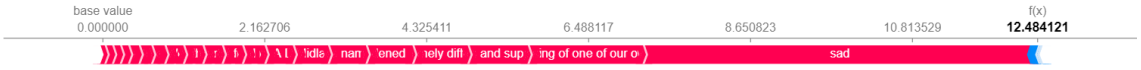
Figure 8: Examples of Negative Emotions in News

(a) Disappointment



CHICAGO, Jan 27 (Reuters) Goldman Sachs analyst Nomi Ghez said Thursday she lowered her rating on Kellogg Co. K.N to market perform from market outperform after reported fourth quarter earnings. Kellogg said it earned 34 cents per share before onetime items in the fourth quarter, matching expectations. However, in a telephone interview, Ghez said revenues of 1.59 billion were below her expectations. "Although earnings were in line and seemed very strong, the top line was very disappointing," she said. The stock was up 916 at 2538.

(b) Sadness

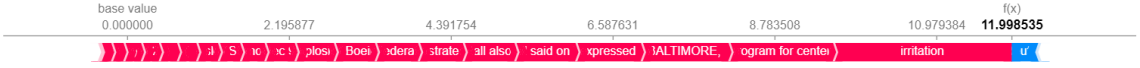


NEW YORK, April 14 (Reuters) A Dow Chemical employee DOW.N died on Thursday after falling inside Midland, Michigan Dow said in a statement. Dow, the largest chemical maker in the United States, declined to release the man's name until his family could be notified. The last fatality at Dow's Michigan was in 1998. "We are greatly saddened by the passing of one of our own," said Earl Shipp, vice president of Dow's Michigan Operations. "This is an extremely difficult time for us, and our deepest sympathies, prayers and support go out to the family and friends." Dow said the worker fell from an "elevated location", but the Midland Daily News reported that he fell 30 feet from a catwalk, citing information from police scanners. Another local newspaper, the Bay City Times, reported that Michigan's Occupational Safety and Health Administration had begun an investigation into the death.

(c) Embarrassment

OMAHA, Neb., March 31 (Reuters) Union Pacific Corp.'s top executive Tuesday said the railroad industry faces a severe capacity problem and apologized for the difficulty and delay has faced in correcting a severe service disruption crisis. "I am acutely embarrassed, and is embarrassed, at the time it has taken to recover from our congestion crisis," said Richard Davidson, chairman and chief executive officer of Union Pacific.

(d) Annoyance



BALTIMORE, Dec 9 (Reuters) National Transportation Safety Board (NTSB) Chairman Jim Hall said on Tuesday he was frustrated by the slow pace of Boeing Co.'s BA.N voluntary inspection program for center fuel tanks on its 747 planes. At a hearing into the explosion of TWA Flight 800, which killed 230 people nearly 17 months ago, Hall also expressed irritation over the Federal Aviation Administration's slow pace in making the Boeing program mandatory for 747s and other airliners.

Note: This table lists four negative emotion examples from the pre-trained BERT model from Demszky et al. (2020). Segments that contribute to positive SHAP values are highlighted in red and segments with negative SHAP values are highlighted in blue.

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Table 13: Cross-Sectional Correlations and Sharpe Ratios of Emotion Portfolios

	Correlations		Sharpe Ratios			Correlations		Sharpe Ratios	
	Mean%	Tstats	EW	VW		Mean%	Tstats	EW	VW
approval	1.71	17.70	1.00	0.23	neutral	-0.43	-4.47	-0.40	-0.21
excitement	1.56	16.40	1.00	0.24	curiosity	0.08	0.84	0.15	0.03
joy	1.31	13.84	0.54	0.29	confusion	-0.00	-0.04	0.05	-0.06
gratitude	1.30	13.55	0.72	0.21	realization	-0.03	-0.31	0.03	0.00
admiration	1.25	12.60	0.74	-0.03	anger	-0.05	-0.58	-0.07	0.03
amusement	0.88	9.39	0.35	-0.10	disgust	-0.08	-0.83	-0.14	-0.10
optimism	0.81	8.41	0.37	0.08	annoyance	-0.14	-1.54	0.10	-0.08
pride	0.77	7.87	0.48	0.08	nervousness	-0.21	-2.17	-0.25	0.02
love	0.70	7.20	0.38	-0.02	disapproval	-0.43	-4.61	0.07	-0.09
surprise	0.60	6.30	0.40	0.03	grief	-0.54	-5.68	-0.46	-0.10
desire	0.51	5.25	0.46	-0.14	remorse	-0.61	-6.41	-0.40	-0.12
relief	0.47	4.94	0.34	0.17	embarrassment	-0.63	-6.59	-0.25	-0.03
fear	0.37	3.89	0.36	0.03	sadness	-0.98	-10.31	-0.72	-0.20
caring	0.10	1.07	-0.07	-0.08	disappointment	-1.20	-12.56	-0.53	-0.26

Note: This table shows the cross-sectional rank correlation between the predicted emotion scores and 3-day returns. We also report the equal-weighted and value-weighted Sharpe ratios of quintile long-short portfolios based on the 28 emotion scores predicted by the pre-trained GoEmotion model from Demszky et al. (2020). The news are from 1996 to 2019. The table is sorted in descending order of t-stats of correlations.

3.10.1 Most Impactful Words

We specifically focus on the SESTM model when identifying the most impactful words, primarily due to its explicit and transparent nature in textual analysis. Figure 9 plots the list of sentiment-charged words estimated by SESTM for US equity market. These are the words that most strongly correlate with realized price fluctuations and thus surpass the correlation screening threshold. Because we re-estimate the model in each of our training samples, the sentiment word lists can change throughout our analysis. The font size of a word in the cloud is proportional to the number of times they appear over all training samples. For English words based on US market, we have positive words such as "raise", "gain", "approval", "strong", "outstanding", which are most often positively correlated to realized returns. On the negative side, we find "fall", "low", "loss", "charge" that are negatively correlated. The appendix provides additional plots for each international market we estimate. A unique advantage of the SESTM approach is that it does not require a language specific LLM. The approach is end-to-end, and automatically adapts to any language.

3.10.2 When Different Models Disagree?

The previous findings have demonstrated that LLMs, such as BERT, outperforms word-based approaches in sentiment analysis. To further understand the underlying differences, we investigate news articles where BERT's sentiment scores diverge from those of Word2vec and SESTM.

To gain insights into model predictions, we employ the SHAP (SHapley Additive exPlanations) method developed by Lundberg and Lee (2017). SHAP is a game-theoretic approach that provides interpretability for machine learning models. SHAP values represent the expected change in predictions based on specific features. For BERT interpretation, we highlight segments contributing to

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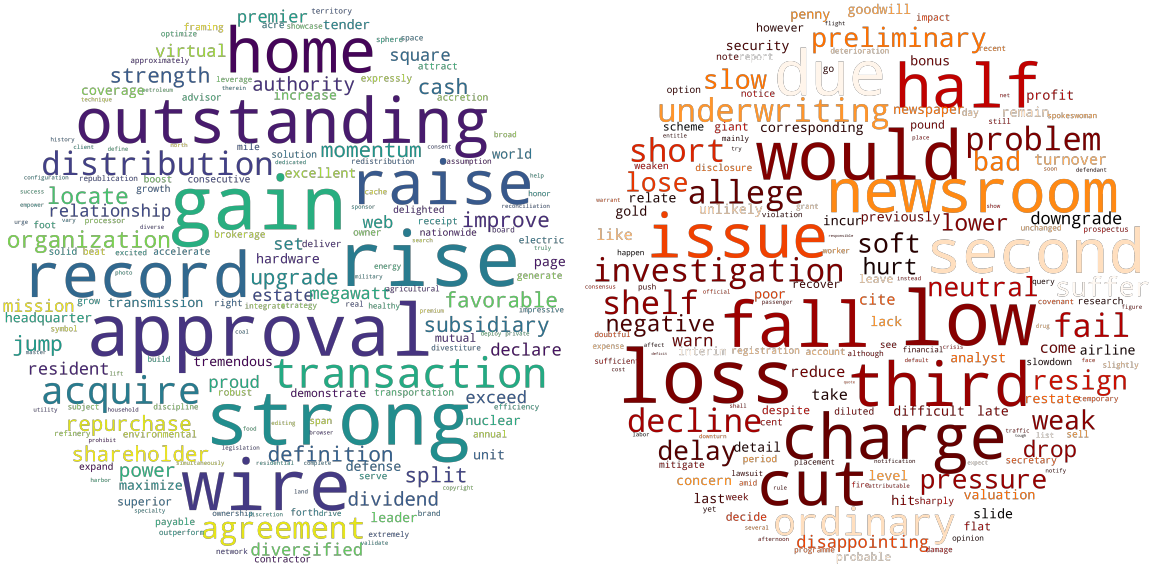
Figure 9: Word Clouds for US



Note: Word cloud for US. Left figure is the word cloud of positive words, right is the word cloud of negative words.

positive SHAP values in red, while segments with negative SHAP values are highlighted in blue. The intensity of the color indicates the magnitude of the SHAP value. For Word2vec and SESTM, we utilize a waterfall plot that highlights the SHAP values for all features, with blue features pushing predictions towards the negative side and red features towards the positive side.

Figure 10 presents an example that illustrates the disagreement among different methods. The accompanying text reads as follows:

Brussels has warned British Airways owner IAG ICAG.L that its favored strategy to allow it to continue flying freely in and around Europe in the event of a no deal Brexit will not work, the Financial Times reported on Tuesday. After Brexit, European carriers will have to show they are more than 50 per cent EU owned and controlled to retain flying rights in the bloc, the FT said. IAG, which also owns the Spanish flag carrier Iberia, is registered in Spain but headquartered in the United Kingdom and has diverse global shareholders. The FT said part of IAG's strategy to retain both EU and UK operating rights is to stress that its important individual airlines are domestically owned through a series of trusts rather than being part of the bigger a high proportion of nonEU investors. The FT quoted an unnamed senior EU official as saying, "For IAG, I can't see how it can be a solution." Concerns have been raised with IAG over its post-Brexit ownership structure, the FT quoted a second Brussels official familiar with the conversations as saying. IAG was not immediately available.

Upon analyzing this news article, it becomes apparent that the message conveyed is negative for British Airways. However, both Word2vec and SESTM label this news as positive, primarily due to the presence of the sentiment word "rise," which is often associated with positive returns. By

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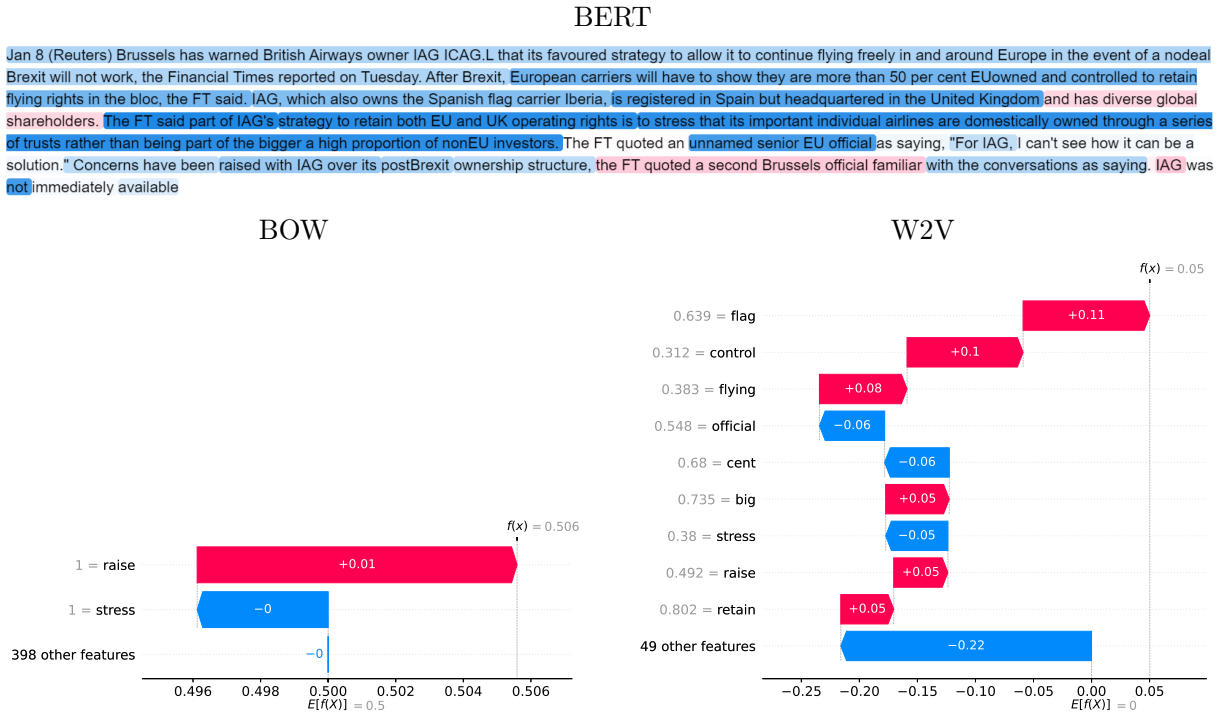
considering the contextual information provided, such as the phrase "concerns have been raised" and the overall content of the paragraph, BERT accurately identifies the sentiment as negative.

This example highlights the importance of analyzing the context surrounding sentiment words and demonstrates BERT's ability to capture nuanced sentiment by considering the entire article.

Figure 10: Disagreement between BERT and Word-based Approaches



Note: This figure includes a piece of news for which BERT labels as negative, whereas Word2vec and SESTM both recognize as positive. Segments/words that contribute to positive SHAP values are highlighted in red and segments/words with negative SHAP values are highlighted in blue.

3.10.3 Impact of Negation Words

The example above demonstrates the problem with word-based approaches – the ignorance of context can lead to sentiment classification errors. We now consider a specific context – negation and examine errors caused by the presence of negation words systematically.

In order to highlight the disagreement between LLMs and word-based methods, we specifically focus on news articles that employ negation words. To do so, we construct double-sorted long-short portfolios based on news containing negation words and compare them to portfolios constructed from news without negation words. The performance of these relative equal-weighted portfolios, based on sentiment analysis, is presented in Table 14.

Upon analyzing the results, we observe that all three LLMs exhibit higher Sharpe ratios for the partition of news articles containing negation words compared to the partition without negation

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words. Specifically, OPT experiences an increase from 3.67 to 3.86, RoBERTa from 3.44 to 3.80, and BERT from 2.95 to 3.48. This suggests that there is greater profitability to be found within the partition of news articles that utilize negation words. This can also be confirmed from the Sharpe ratios earned from shorting side of these portfolios.

On the contrary, the word-based models demonstrate even worse portfolio performance when negation words are present. Word2vec's performance decreases from 2.79 to 1.94, while SESTM's performance decreases from 3.29 to 2.85. It becomes apparent that word-based models tend to make more errors or misinterpretations when handling news articles that contain negation words. In contrast, LLMs are able to effectively leverage the presence of negation words to identify hidden predictive signals or seize additional opportunities in the data.

To quantitatively assess the influence of negation words on the performance of word-based models, we further undertake a regression to delve into the impact of negation words on word-based models' performance compared to LLMs as follows:

$$r_{h,i,t+1}(s_{i,t}^{LLM} - s_{i,t}^{\text{word-based model}}) = \alpha + \beta \text{Negation}_{i,t} + \text{Fixed Effect} + \text{Control Variables}_{i,t} + \epsilon_{i,t}$$

For the left hand side of the regression, our primary focus is on the return-weighted difference between LLM signals and word-based signals. This choice of dependent variable is essential as it enables us to account for the differential signals and their correlation with actual returns, which could also be interpreted as the difference in bets weighted by the success of the bets. We still employ Word2Vec and SESTM as benchmark comparators.

For the right hand side of the regression, we employ two distinct measures to capture the essence of negation words in our analysis: negation word count and negation word ratio. These measures serve as proxies for the presence and prevalence of negation words in the text. Additionally, we introduce several classic factors as control variables including the Fama-French three factors to better determine the impact of negation words. For additional robustness, we also report results in the appendix, augmenting the model with momentum and industry factors. Moreover, we integrate text feature-related variables as controls to enhance the precision of our analysis.

The results of the regression is displayed in Table 15 and Table 16. From these tables, we can see that both negation word count and negation word ratio exhibit significant, positive coefficients. Specifically, this effect is accentuated in models that typically exhibit superior performance. This signifies that, when negation words are present, LLMs tend to outperform word-based models in their predictions.

These findings underscore the advantage of LLMs over word-based models when it comes to capturing the nuances and subtleties within news articles, particularly in the context of negation. While word-based models struggle with the presence of negation words, LLMs demonstrate their ability to harness the underlying information and exploit the predictive signals that may be concealed within such articles. This distinction further highlights the superior performance and interpretability

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of LLMs in sentiment analysis tasks.

Table 14: Portfolio Performance Comparison on News With and Without Negation Words

	OPT						RoBERTa					
	W/O Negation Words			W/ Negation Words			W/O Negation Words			W/ Negation Words		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.41	-0.07	0.48	0.46	-0.19	0.65	0.39	-0.05	0.43	0.48	-0.15	0.62
Std	0.21	0.24	0.13	0.22	0.25	0.17	0.21	0.23	0.13	0.22	0.25	0.16
SR	1.99	-0.27	3.67	2.13	-0.73	3.86	1.87	-0.21	3.44	2.17	-0.59	3.80
	BERT						Word2vec					
	W/O Negation Words			W/ Negation Words			W/O Negation Words			W/ Negation Words		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.34	-0.01	0.35	0.47	-0.10	0.58	0.31	-0.02	0.33	0.34	0.05	0.29
Std	0.21	0.23	0.12	0.22	0.24	0.17	0.21	0.23	0.12	0.22	0.24	0.15
SR	1.65	-0.04	2.95	2.16	-0.43	3.48	1.51	-0.08	2.79	1.59	0.23	1.94
	SESTM						FinBERT					
	W/O Negation Words			W/ Negation Words			W/O Negation Words			W/ Negation Words		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.35	-0.05	0.41	0.40	-0.02	0.43	0.33	0.17	0.16	0.36	0.18	0.18
Std	0.21	0.23	0.12	0.22	0.24	0.15	0.21	0.20	0.11	0.23	0.21	0.15
SR	1.69	-0.23	3.29	1.87	-0.09	2.85	1.54	0.85	1.43	1.59	0.83	1.20

Note: This table compares the long-short portfolio performance of news articles with or without negation words built on the basis of OPT, RoBERTa, BERT, Word2vec, SESTM, and FinBERT models, respectively. On each day, we filter the news with or without negation words (followed by sentiment words identified by SESTM) and long/short the top/bottom half of stocks based on the sentiment scores, respectively.

3.11 Impact of Truncation

Our Large Language Model employs a strategy of partial token integration, which inherently leads to truncation for longer sequences. Consequently, certain lengthier sequences undergo token truncation, potentially resulting in the loss of valuable information. This limitation arises from the nature of our LLM's design, which processes only a subset of the complete token sequence, inadvertently excluding crucial information embedded within the truncated tokens.

A distinguishing feature of our OPT model is its capacity to process up to 1024 tokens, a notable departure from other Large Language Models constrained to a maximum of 512 tokens. Utilizing the OPT tokenizer, we find that 57.8% of the articles contain fewer than 512 tokens, while an additional 26% of the articles have token counts falling between 512 and 1024. With this expanded token capacity for each article, we can investigate the impact of truncation, specifically whether introducing additional tokens to the model enhances prediction accuracy.

To delve into this, we conduct a comparative analysis, pitting the performance of OPT models using 512 tokens against those utilizing 1024 tokens. It's worth highlighting that our word-based model operates without token-based constraints, allowing us to simultaneously explore performance disparities between word-based and OPT models in terms of token quantity processed.

We introduce a methodological framework for categorizing articles based on token count into three distinct classes: those with fewer than 512 tokens, those encompassing tokens ranging from

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Table 15: Impact of negation word count with FF3 factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words count	0.0079***	0.0040***	0.0003	0.0107*	0.0119***	0.0080***	0.0043***	0.0147**
	(0.0008)	(0.0008)	(0.0007)	(0.0061)	(0.0007)	(0.0007)	(0.0005)	(0.0061)
beta	0.0031***	0.0033***	0.0010	0.0150*	0.0023**	0.0024**	0.0002	0.0142
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0509***	-0.0493***	-0.0596***	0.2831***	-0.0048**	-0.0032*	-0.0136***	0.3291***
	(0.0023)	(0.0023)	(0.0019)	(0.0174)	(0.0020)	(0.0019)	(0.0014)	(0.0176)
bm	0.0013	0.0014	0.0017*	-0.0075	-0.0003	-0.0002	0.0000	-0.0091
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0009)	(0.0007)	(0.0087)
Constant	0.0609***	0.0556***	0.0509***	-0.0425***	0.0277***	0.0224***	0.0177***	-0.0757***
	(0.0010)	(0.0010)	(0.0008)	(0.0077)	(0.0009)	(0.0009)	(0.0006)	(0.0078)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0316	0.0262	0.0369	0.0348	0.0193	0.0170	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word count on the difference between LLM and word-based model performance with FF3 factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

Table 16: Impact of negation word ratio with FF3 factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words ratio	0.0069***	0.0034***	-0.0001	0.0109*	0.0113***	0.0078***	0.0042***	0.0153***
	(0.0008)	(0.0008)	(0.0006)	(0.0058)	(0.0007)	(0.0006)	(0.0005)	(0.0059)
beta	0.0031***	0.0033***	0.0010	0.0150*	0.0023**	0.0024**	0.0002	0.0142
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0510***	-0.0493***	-0.0597***	0.2830***	-0.0049**	-0.0032*	-0.0136***	0.3291***
	(0.0023)	(0.0023)	(0.0019)	(0.0174)	(0.0020)	(0.0019)	(0.0014)	(0.0176)
bm	0.0013	0.0014	0.0017*	-0.0075	-0.0003	-0.0002	0.0000	-0.0091
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0009)	(0.0007)	(0.0087)
Constant	0.0609***	0.0556***	0.0509***	-0.0425***	0.0277***	0.0224***	0.0177***	-0.0756***
	(0.0010)	(0.0010)	(0.0008)	(0.0077)	(0.0009)	(0.0009)	(0.0006)	(0.0078)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0315	0.0262	0.0369	0.0348	0.0193	0.0170	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word ratio on the difference between LLM and word-based model performance with FF3 factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table 17: Performance Comparison with Different Token Truncation

	OPT		OPTTK1024		SESTM		Word2vec	
	Acc.	Corr.	Acc.	Corr.	Acc.	Corr.	Acc.	Corr.
token length $\leq$ 512	54.37	1.58	54.42	1.60	52.37	2.16	54.00	1.58
512 $<$ token length $<$ 1024	54.02	1.77	54.07	1.75	52.07	1.70	53.89	1.66
token length $\geq$ 1024	54.03	2.86	53.99	2.82	51.94	1.71	53.44	2.55
Overall	54.26	1.99	54.31	2.01	52.24	2.02	53.89	1.86
	Return	Sharpe Ratio	Return	Sharpe Ratio	Return	Sharpe Ratio	Return	Sharpe Rat
Sentiment	48.73	4.51	50.85	4.78	35.33	3.61	30.16	3.13
Prediction	50.18	4.84	50.33	4.79			29.40	2.63

Note: This table reports the performance of OPT, OPTTK1024, SESTM, and Word2vec models with different token number. The upper part of the table reports the sentiment classification accuracy and cross-sectional prediction correlation of the models with different token number. The lower part of the table reports the portfolio performance of the models.

512 to 1024, and those surpassing 1024 tokens. This classification system assigns token types as 1, 2, and 3 respectively.

We begin by presenting the performance report in Table 17, which assesses the performance of the three models across various token types. We provide the results both for sentiment classification accuracy and cross-sectional prediction correlation. Additionally, we report the performance of portfolios constructed based on each respective model in the lower sections of the table. Notably, the data indicates that OPTTK1024 generally outperforms the other models. However, it's worth noting that OPTTK1024 doesn't exhibit significant superiority for articles with more than 1024 tokens.

In order to explore the potential impact of token truncation on the observed disparities, we delve into the correlation between the differences in model signals and the token type variable. Similar to the approach outlined in section 3.10.3, we continue to employ the measure of model discrepancies as the difference in signals multiplied by the next period return. We then regress these discrepancies against token type while also considering the inherent characteristics of individual stocks. These characteristics include the Fama-French 3 factors, momentum, and industry factors, resulting in the following equation:

$$\text{Model Signal Difference}_{i,t} = \alpha + \beta \text{Token Type} i,t + \text{Fixed Effect} + \text{Control Variables}_{i,t} + \epsilon_{i,t}$$

The regression results are presented in Tables 18 to 20. Tables 18 and 19 display the regressions of the difference between OPT and OPTTK1024 with SESTM on token type. Table 20, on the other hand, focuses on the comparison between OPT and OPTTK1024 models. Additionally, we conducted an examination using Word2Vec as an alternative benchmark for word-based models, with the results provided in the appendix.

Tables 18 and 19 reveal that both OPT and OPTTK1024 exhibit superiority over word-based models when the number of tokens is large. However, although OPTTK1024 has a larger coefficient

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in magnitude for the "token type" variable compared to the OPT model, Table 4 indicates that when regressing their differences, the coefficient of token type is positive but not statistically significant. This suggests that while OPTTK1024 generally outperforms the OPT model in prediction and performance, it does not play a statistically significant role. Thus, the use of the OPT model with 512 tokens is deemed appropriate for model comparisons throughout the remainder of the paper.

4 Conclusion

In this study, we have harnessed the power of state-of-the-art LLMs in NLP to obtain contextualized embeddings of news text. Our comprehensive analysis has encompassed a wide range of news articles from 16 countries, written in 13 different languages, providing compelling evidence of the predictability of returns based on news data.

One key advantage of LLMs is their ability to capture and incorporate contextual information, surpassing the limitations of traditional word-based approaches. Unlike methods that rely solely on the frequency of individual words or simplistic word vectors with limited context, our utilization of LLMs allows for a more nuanced understanding of sentiment and return prediction. This heightened accuracy in capturing the underlying statistical patterns translates into superior economic performance, as demonstrated by the impressive Sharpe ratios of portfolios constructed using sentiment scores and predicted returns.

The robustness of our results across multiple countries and languages underscores the generalizability and applicability of LLMs in diverse financial contexts. Furthermore, our study serves as a catalyst for further exploration, encouraging innovative approaches that integrate textual analysis with empirical finance. This intersection opens exciting avenues for understanding and effectively utilizing textual data to gain an edge in investment strategies. With our research serving as a starting point, we anticipate exciting developments and discoveries in this fascinating area of study.

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Table 18: Difference in OPTTK1024 and SESTM

	FF3	FF3+Momentum				FF3+Industry
token type	0.0252***	0.0252***	0.0252***	0.0252***	0.0235***	0.0252***
	(0.0017)	(0.0017)	(0.0017)	(0.0017)	(0.0018)	(0.0017)
beta	0.0039***	0.0039***	0.0035***	0.0033***	0.0026**	0.0039***
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)
mvel1	-0.0511***	-0.0507***	-0.0459***	-0.0437***	-0.0502***	-0.0515***
	(0.0023)	(0.0023)	(0.0023)	(0.0024)	(0.0025)	(0.0023)
bm	0.0015	0.0015	0.0020*	0.0019*	0.0019	0.0014
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0012)	(0.0011)
mom1m		-0.0013*				
		(0.0007)				
mom6m			-0.0079***			
			(0.0007)			
mom12m				-0.0083***		
				(0.0007)		
mom36m					-0.0002	
					(0.0008)	
sic2						0.0004**
						(0.0001)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,324,399	1,324,399	1,324,399	1,324,377	1,236,214	1,324,399
Adj R-squared	0.0290	0.0290	0.0291	0.0291	0.0302	0.0290

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of truncation in token length on the difference between OPTTK1024 model and SESTM performance. We use OPTTK1024 signals minus SESTM signals multiplied by next period return as the dependent variable. We include Fama-French 3 factors, momentum factors, and industry factors and other controls as independent variables. The sample period is from 2004 to 2019.

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Table 19: Difference in OPT and SESTM

	FF3	FF3+Momentum				FF3+Industry
token type	0.0247***	0.0247***	0.0248***	0.0248***	0.0231***	0.0247***
	(0.0018)	(0.0018)	(0.0018)	(0.0018)	(0.0018)	(0.0018)
beta	0.0038***	0.0038***	0.0035***	0.0032***	0.0027**	0.0039***
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)
mvel1	-0.0543***	-0.0540***	-0.0491***	-0.0471***	-0.0537***	-0.0547***
	(0.0023)	(0.0023)	(0.0024)	(0.0024)	(0.0025)	(0.0023)
bm	0.0011	0.0011	0.0016	0.0015	0.0017	0.0010
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)
mom1m		-0.0010				
		(0.0007)				
mom6m			-0.0079***			
			(0.0007)			
mom12m				-0.0081***		
				(0.0007)		
mom36m					-0.0001	
					(0.0008)	
sic2						0.0003**
						(0.0001)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,324,399	1,324,399	1,324,399	1,324,377	1,236,214	1,324,399
Adj R-squared	0.0294	0.0294	0.0295	0.0295	0.0305	0.0294

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of truncation in token length on the difference between OPT model and SESTM performance. We use OPT signals minus SESTM signals multiplied by next period return as the dependent variable. We include Fama-French 3 factors, momentum factors, and industry factors and other controls as independent variables. The sample period is from 2004 to 2019.

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Table 20: Difference in OPT and OPTTK1024

	FF3	FF3+Momentum				FF3+Industry
token type	0.0005	0.0005	0.0005	0.0005	0.0004	0.0005
	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)
beta	0.0000	0.0000	0.0001	0.0000	-0.0001	0.0001
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
mvel1	0.0033***	0.0033***	0.0032***	0.0034***	0.0034***	0.0032***
	(0.0006)	(0.0006)	(0.0006)	(0.0006)	(0.0007)	(0.0006)
bm	0.0004	0.0004	0.0004	0.0004	0.0002	0.0004
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
mom1m		-0.0002				
		(0.0002)				
mom6m			0.0001			
			(0.0002)			
mom12m				-0.0001		
				(0.0002)		
mom36m					-0.0001	
					(0.0002)	
sic2						0.0000
						(0.0000)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,324,399	1,324,399	1,324,399	1,324,377	1,236,214	1,324,399
Adj R-squared	0.0071	0.0071	0.0071	0.0071	0.0078	0.0071

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of truncation in token length on the difference between OPTTK1024 model and OPT performance. We use OPTTK1024 signals minus OPT signals multiplied by next period return as the dependent variable. We include Fama-French 3 factors, momentum factors, and industry factors and other controls as independent variables. The sample period is from 2004 to 2019.

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Appendix A Additional Tables and Figures

Table IA1: International News Summary Statistics

	Raw Articles			Articles Tagged with Single Stock			Articles With Returns	After Filtering Short Articles	After Filtering Similarity > 0.8
	RTRS	3PTY	Total	RTRS	3PTY	Total			
UK	707,288	1,050,467	1,757,755	196,573	773,266	969,839	906,705	901,838	571,285
Australia	261,020	1,203,784	1,464,804	100,444	1,113,347	1,213,791	388,585	382,114	249,190
Canada	255,933	473,686	729,619	126,281	431,401	557,682	481,891	478,205	350,549
China (HK)	3,537,487	7,287,688	10,825,175	1,140,542	5,558,763	6,699,305	2,086,045	305,335	182,363
Japan	3,259,103	38,860	3,297,963	1,210,077	16,850	1,226,927	405,341	399,185	310,244
Germany	2,423,671	1,751,231	4,174,902	480,264	880,650	1,360,914	238,577	229,265	178,039
Italy	1,022,204	337,322	1,359,526	194,650	227,599	422,249	173,250	168,410	130,168
France	2,422,338	1,587,490	4,009,828	298,886	670,469	969,355	174,917	174,784	153,779
Sweden	288,395	189,424	477,819	96,039	124,862	220,901	126,211	126,168	115,195
Denmark	261,146	124,209	385,355	93,596	57,768	151,364	53,056	52,381	43,584
Spain	2,748,601	165,468	2,914,069	257,739	46,829	304,568	47,541	45,597	34,159
Finland	81	110,123	110,204	38	87,226	87,264	38,159	38,119	28,633
Portugal	747,069	39,086	786,155	124,017	13,638	137,655	11,265	11,212	6,158
Greece	85,915	14	85,929	19,156	6	19,162	10,093	10,082	7,710
Netherlands	194	183,668	183,862	53	66,669	66,722	4,313	4,312	3,751

Note: In this table, we report the remaining sample size after each filter applied on the news articles for international equity markets. Column "Raw Articles" presents the numbers of available articles separately from Thomson Reuters Real-time News Feed (RTRS) and Archive (3PTY). Columns under "Articles Tagged with Single Stock" presents the number of articles tagged with a single stock. Column "Articles with Available Returns" presents the number of remaining articles after matching returns data. Column "After Filtering Short Articles" reports the number of articles with at least 100 characters and at most 100,000 characters.

Figure IA1: Word Clouds for China (HK)

Note: Word cloud for China (HK). Left figure is the word cloud of positive words, right is the word cloud of negative words.

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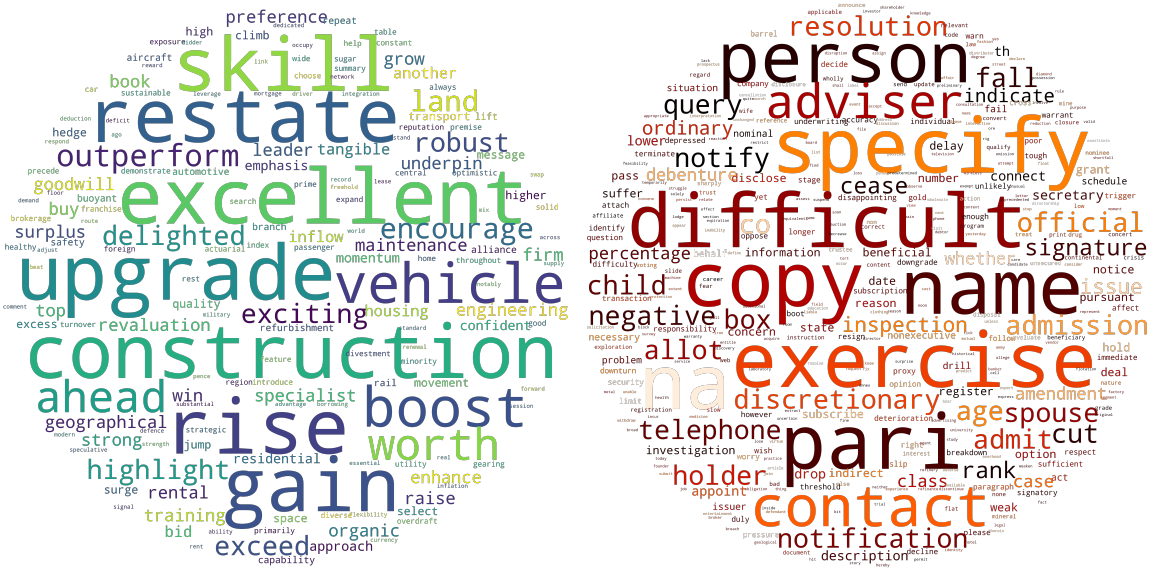
Table IA2: Specification of Tokenizers

	BOW/Word Embeddings	BERT	RoBERTa
US	en_core_web_sm	bert-large-uncased	roberta-large
UK	en_core_web_sm	bert-large-uncased	roberta-large
Australia	en_core_web_sm	bert-large-uncased	roberta-large
Canada	en_core_web_sm	bert-large-uncased	roberta-large
China (HK)	jieba	ckiplab/bert-base-chinese	xlm-roberta-large
Japan	nagisa	cl-tohoku/bert-base-japanese	xlm-roberta-large
Germany	de_core_news_sm	bert-base-german-cased	xlm-roberta-large
Italy	it_core_news_sm	dbmdz/bert-base-italian-cased	xlm-roberta-large
France	fr_core_news_sm	dbmdz/bert-base-french-europeana-cased	xlm-roberta-large
Sweden	xx_ent_wiki_sm	KB/bert-base-swedish-cased	xlm-roberta-large
Denmark	da_core_news_sm	Maltehb/danish-bert-botxo	xlm-roberta-large
Spain	es_core_news_sm	dccuchile/bert-base-spanish-wwm-uncased	xlm-roberta-large
Finland	xx_ent_wiki_sm	TurkuNLP/bert-base-finnish-cased-v1	xlm-roberta-large
Portugal	pt_core_news_sm	neuralmind/bert-base-portuguese-cased	xlm-roberta-large
Greece	el_core_news_sm	nlpaueb/bert-base-greek-uncased-v1	xlm-roberta-large
Netherlands	nl_core_news_sm	wietsedv/bert-base-dutch-cased	xlm-roberta-large

Note: This table reports the specification of the pre-trained model for each country. Column "BOW/Word Embeddings" reports the specific tokenizer in spaCy that we use. Column "BERT" and "RoBERTa" reports pre-trained models for BERT and RoBERTa from Hugging Face.

Figure IA2: Word Clouds for UK



Note: Word cloud for UK. Left figure is the word cloud of positive words, right is the word cloud of negative words.

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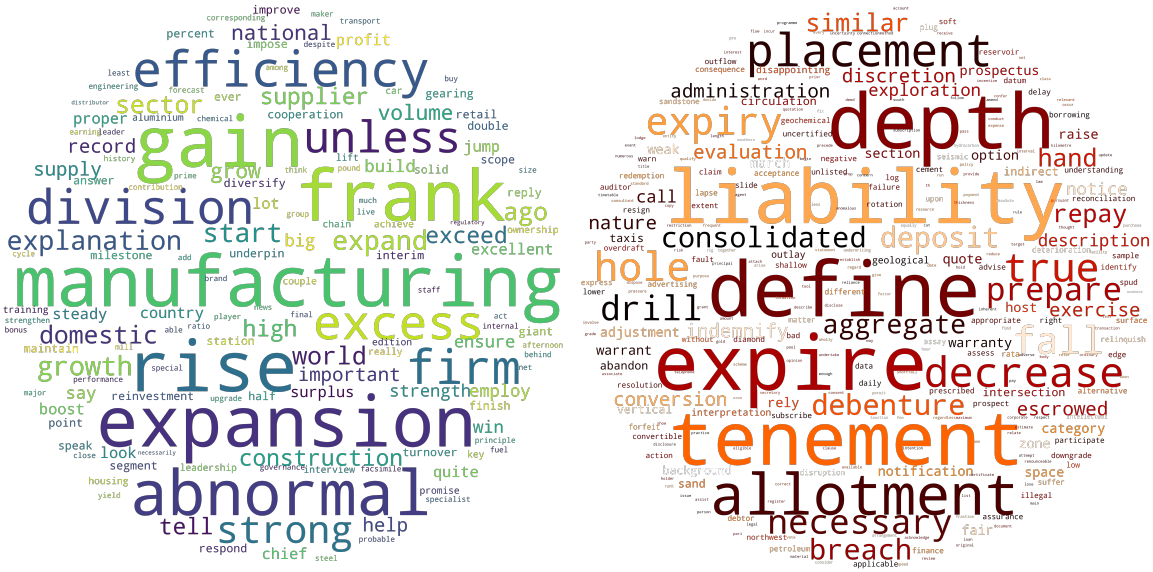
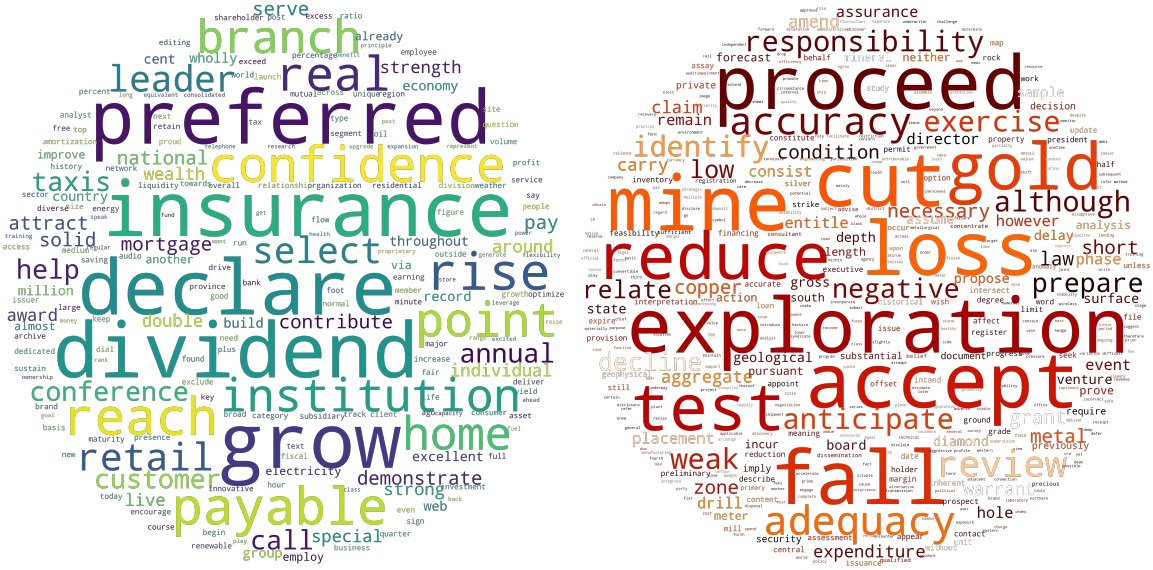
Figure IA3: Word Clouds for Australia



Note: Word cloud for Australia. Left figure is the word cloud of positive words, right is the word cloud of negative words.

Figure IA4: Word Clouds for Canada



Note: Word cloud for Canada. Left figure is the word cloud of positive words, right is the word cloud of negative words.

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Figure IA5: Word Clouds for Japan

Note: Word cloud for Japan. Left figure is the word cloud of positive words, right is the word cloud of negative words.

Figure IA6: Word Clouds for Germany

Note: Word cloud for Germany. Left figure is the word cloud of positive words, right is the word cloud of negative words.

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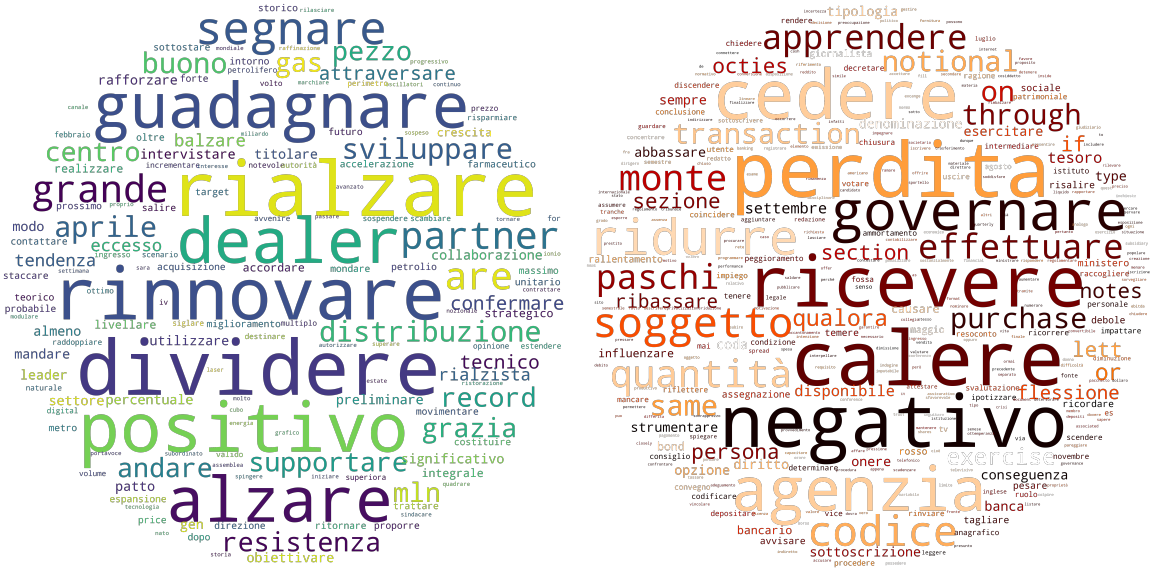
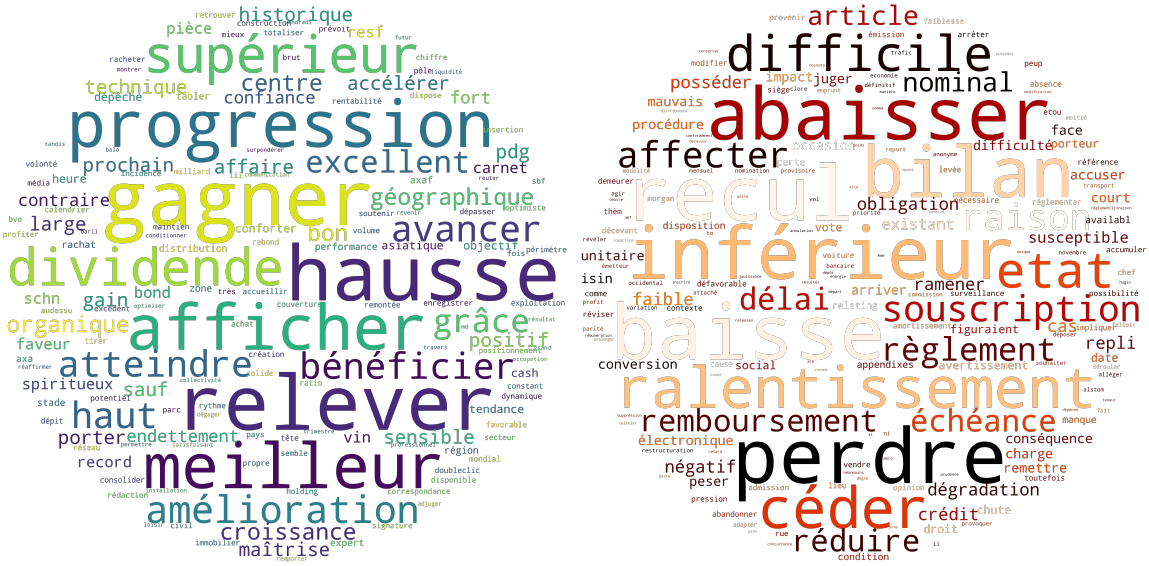
Figure IA7: Word Clouds for Italy



Note: Word cloud for Italy. Left figure is the word cloud of positive words, right is the word cloud of negative words.

Figure IA8: Word Clouds for France



Note: Word cloud for France. Left figure is the word cloud of positive words, right is the word cloud of negative words.

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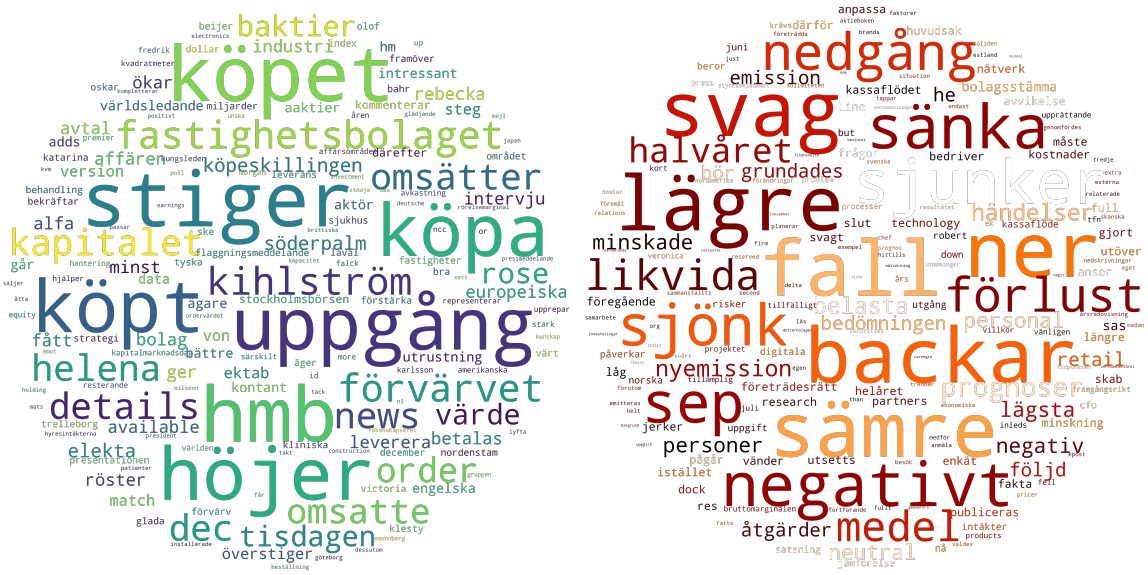
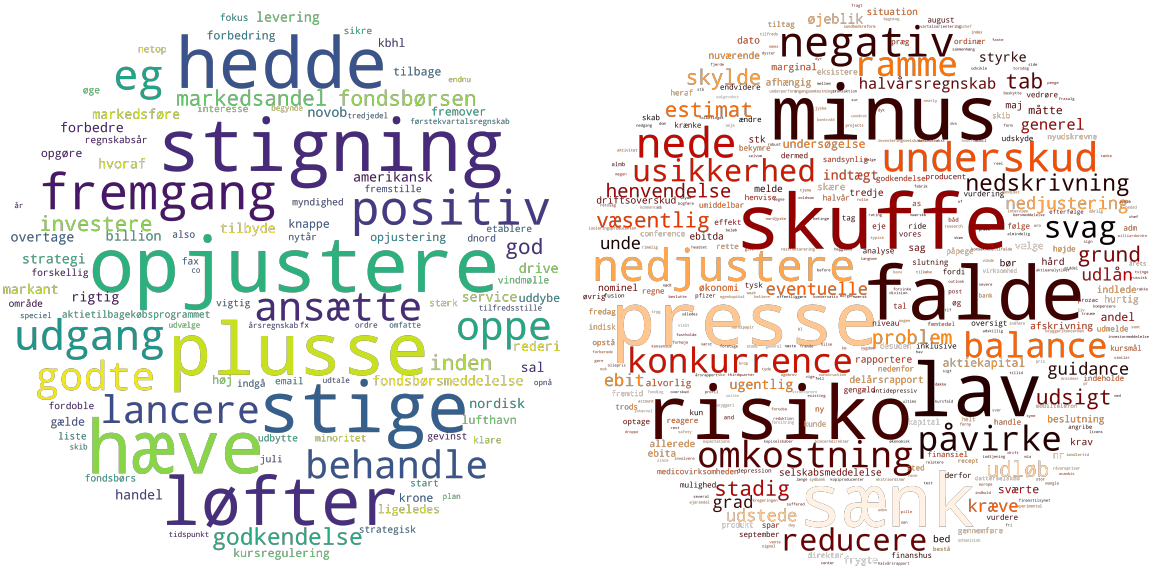
Figure IA9: Word Clouds for Sweden



Note: Word cloud for Sweden. Left figure is the word cloud of positive words, right is the word cloud of negative words.

Figure IA10: Word Clouds for Denmark



Note: Word cloud for Denmark. Left figure is the word cloud of positive words, right is the word cloud of negative words.

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Figure IA11: Word Clouds for Spain

Note: Word cloud for Spain. Left figure is the word cloud of positive words, right is the word cloud of negative words.

Figure IA12: Word Clouds for Finland

Note: Word cloud for Finland. Left figure is the word cloud of positive words, right is the word cloud of negative words.

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Figure IA13: Word Clouds for Portugal

Note: Word cloud for Portugal. Left figure is the word cloud of positive words, right is the word cloud of negative words.

Figure IA14: Word Clouds for Greece

Note: Word cloud for Greece. Left figure is the word cloud of positive words, right is the word cloud of negative words.

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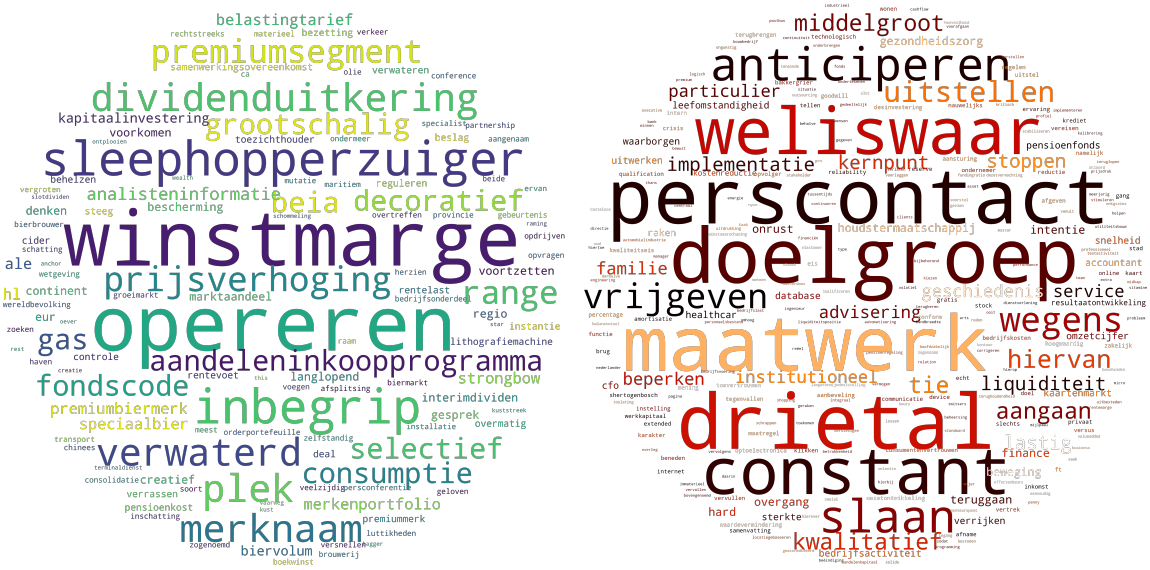
Figure IA15: Word Clouds for Netherlands



Note: Word cloud for Netherlands. Left figure is the word cloud of positive words, right is the word cloud of negative words.

Table IA3: Impact of negation word count with FF3 and mom1m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words count	0.0079***	0.0040***	0.0003	0.0107*	0.0119***	0.0080***	0.0043***	0.0147**
	(0.0008)	(0.0008)	(0.0007)	(0.0061)	(0.0007)	(0.0007)	(0.0005)	(0.0061)
beta	0.0031***	0.0033***	0.0010	0.0150*	0.0023**	0.0024**	0.0002	0.0142
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0506***	-0.0488***	-0.0592***	0.2862***	-0.0045**	-0.0028	-0.0132***	0.3322***
	(0.0023)	(0.0023)	(0.0019)	(0.0175)	(0.0020)	(0.0019)	(0.0014)	(0.0176)
bm	0.0014	0.0015	0.0017*	-0.0073	-0.0003	-0.0002	0.0001	-0.0089
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0009)	(0.0007)	(0.0087)
mom1m	-0.0011	-0.0015**	-0.0013**	-0.0094*	-0.0009	-0.0013**	-0.0011***	-0.0092*
Constant	0.0607***	0.0554***	0.0507***	-0.0435***	0.0276***	0.0223***	0.0176***	-0.0767***
	(0.0010)	(0.0010)	(0.0008)	(0.0077)	(0.0009)	(0.0009)	(0.0006)	(0.0078)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0316	0.0262	0.0369	0.0348	0.0193	0.0170	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word count on the difference between LLM and word-based model performance with FF3 and mom1m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA4: Impact of negation word count with FF3 and mom6m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words count	0.0079***	0.0040***	0.0002	0.0107*	0.0119***	0.0080***	0.0043***	0.0147**
	(0.0008)	(0.0008)	(0.0007)	(0.0061)	(0.0007)	(0.0007)	(0.0005)	(0.0061)
beta	0.0028**	0.0030**	0.0008	0.0146*	0.0021**	0.0023**	0.0001	0.0139
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0461***	-0.0450***	-0.0563***	0.2885***	-0.0018	-0.0007	-0.0120***	0.3327***
	(0.0024)	(0.0024)	(0.0019)	(0.0177)	(0.0020)	(0.0020)	(0.0014)	(0.0179)
bm	0.0019	0.0019	0.0020**	-0.0069	0.0000	0.0001	0.0002	-0.0088
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0087)
mom6m	-0.0074***	-0.0065***	-0.0051***	-0.0082	-0.0046***	-0.0037***	-0.0023***	-0.0054
Constant	0.0593***	0.0542***	0.0498***	-0.0442***	0.0267***	0.0216***	0.0172***	-0.0768***
	(0.0010)	(0.0010)	(0.0008)	(0.0078)	(0.0009)	(0.0009)	(0.0006)	(0.0079)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0316	0.0263	0.0369	0.0348	0.0193	0.0171	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word count on the difference between LLM and word-based model performance with FF3 and mom6m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA5: Impact of negation word count with FF3 and mom12m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words count	0.0078***	0.0040***	0.0002	0.0106*	0.0119***	0.0080***	0.0043***	0.0146**
	(0.0008)	(0.0008)	(0.0007)	(0.0061)	(0.0007)	(0.0007)	(0.0005)	(0.0061)
beta	0.0026**	0.0028**	0.0006	0.0141	0.0020**	0.0022**	0.0000	0.0135
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0441***	-0.0432***	-0.0549***	0.2947***	-0.0005	0.0005	-0.0113***	0.3384***
	(0.0024)	(0.0024)	(0.0019)	(0.0181)	(0.0020)	(0.0020)	(0.0015)	(0.0183)
bm	0.0017	0.0018	0.0019**	-0.0069	-0.0000	0.0000	0.0002	-0.0086
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0087)
mom12m	-0.0076***	-0.0069***	-0.0053***	-0.0130**	-0.0048***	-0.0041***	-0.0025***	-0.0102*
Constant	0.0586***	0.0536***	0.0493***	-0.0463***	0.0263***	0.0212***	0.0169***	-0.0787***
	(0.0011)	(0.0010)	(0.0008)	(0.0079)	(0.0009)	(0.0009)	(0.0006)	(0.0080)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473
Adj R-squared	0.0316	0.0263	0.0369	0.0348	0.0193	0.0171	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word count on the difference between LLM and word-based model performance with FF3 and mom12m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA6: Impact of negation word count with FF3 and mom36m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words count	0.0068***	0.0033***	-0.0002	0.0072	0.0108***	0.0073***	0.0038***	0.0111*
	(0.0008)	(0.0008)	(0.0007)	(0.0061)	(0.0007)	(0.0007)	(0.0005)	(0.0062)
beta	0.0022*	0.0027**	-0.0004	0.0218**	0.0023**	0.0028***	-0.0003	0.0218**
	(0.0012)	(0.0012)	(0.0010)	(0.0091)	(0.0010)	(0.0010)	(0.0007)	(0.0092)
mvel1	-0.0499***	-0.0480***	-0.0588***	0.2671***	-0.0029	-0.0010	-0.0118***	0.3140***
	(0.0025)	(0.0025)	(0.0020)	(0.0189)	(0.0021)	(0.0021)	(0.0016)	(0.0191)
bm	0.0022*	0.0021*	0.0021**	-0.0077	0.0008	0.0007	0.0006	-0.0091
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mom36m	0.0005	0.0004	-0.0007	0.0068	0.0006	0.0006	-0.0005	0.0069
Constant	0.0598***	0.0545***	0.0509***	-0.0466***	0.0257***	0.0205***	0.0168***	-0.0806***
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0086)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470
Adj R-squared	0.0323	0.0269	0.0381	0.0361	0.0202	0.0175	0.0125	0.0406

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word count on the difference between LLM and word-based model performance with FF3 and mom36m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA7: Impact of negation word count with FF3 and industry factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words count	0.0079***	0.0040***	0.0003	0.0107*	0.0119***	0.0080***	0.0043***	0.0147**
	(0.0008)	(0.0008)	(0.0007)	(0.0061)	(0.0007)	(0.0007)	(0.0005)	(0.0061)
beta	0.0032***	0.0033***	0.0010	0.0149*	0.0023**	0.0024**	0.0002	0.0140
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0512***	-0.0496***	-0.0601***	0.2840***	-0.0049**	-0.0032*	-0.0137***	0.3303***
	(0.0023)	(0.0023)	(0.0019)	(0.0175)	(0.0020)	(0.0019)	(0.0014)	(0.0176)
bm	0.0013	0.0013	0.0016*	-0.0073	-0.0003	-0.0002	-0.0000	-0.0089
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0087)
sic2	0.0002*	0.0003*	0.0004***	-0.0008	0.0000	0.0000	0.0001	-0.0010
Constant	0.0490***	0.0430***	0.0340***	-0.0038	0.0262***	0.0202***	0.0112***	-0.0266
	(0.0071)	(0.0071)	(0.0057)	(0.0532)	(0.0060)	(0.0059)	(0.0043)	(0.0537)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0316	0.0262	0.0369	0.0348	0.0193	0.0170	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word count on the difference between LLM and word-based model performance with FF3 and industry factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA8: Impact of negation word ratio with FF3 and mom1m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words ratio	0.0069***	0.0034***	-0.0001	0.0108*	0.0113***	0.0078***	0.0042***	0.0152***
	(0.0008)	(0.0008)	(0.0006)	(0.0058)	(0.0007)	(0.0006)	(0.0005)	(0.0059)
beta	0.0032***	0.0033***	0.0010	0.0150*	0.0023**	0.0025**	0.0002	0.0142
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0506***	-0.0488***	-0.0592***	0.2861***	-0.0046**	-0.0028	-0.0132***	0.3322***
	(0.0023)	(0.0023)	(0.0019)	(0.0175)	(0.0020)	(0.0019)	(0.0014)	(0.0176)
bm	0.0014	0.0015	0.0017*	-0.0073	-0.0003	-0.0002	0.0001	-0.0089
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0009)	(0.0007)	(0.0087)
mom1m	-0.0011	-0.0015**	-0.0013**	-0.0093*	-0.0009	-0.0013**	-0.0011***	-0.0092*
Constant	0.0608***	0.0554***	0.0507***	-0.0435***	0.0276***	0.0223***	0.0176***	-0.0767***
	(0.0010)	(0.0010)	(0.0008)	(0.0077)	(0.0009)	(0.0009)	(0.0006)	(0.0078)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0315	0.0262	0.0369	0.0348	0.0193	0.0170	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word ratio on the difference between LLM and word-based model performance with FF3 and mom1m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA9: Impact of negation word ratio with FF3 and mom6m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words ratio	0.0069***	0.0033***	-0.0002	0.0108*	0.0113***	0.0077***	0.0042***	0.0152***
	(0.0008)	(0.0008)	(0.0006)	(0.0058)	(0.0007)	(0.0006)	(0.0005)	(0.0059)
beta	0.0028**	0.0030**	0.0008	0.0146*	0.0021**	0.0023**	0.0001	0.0139
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0461***	-0.0450***	-0.0563***	0.2884***	-0.0019	-0.0008	-0.0120***	0.3327***
	(0.0024)	(0.0024)	(0.0019)	(0.0177)	(0.0020)	(0.0020)	(0.0014)	(0.0179)
bm	0.0019	0.0019	0.0020**	-0.0069	0.0000	0.0001	0.0002	-0.0088
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0087)
mom6m	-0.0074***	-0.0065***	-0.0051***	-0.0082	-0.0046***	-0.0037***	-0.0023***	-0.0054
Constant	0.0593***	0.0542***	0.0498***	-0.0442***	0.0267***	0.0216***	0.0172***	-0.0768***
	(0.0010)	(0.0010)	(0.0008)	(0.0078)	(0.0009)	(0.0009)	(0.0006)	(0.0079)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0316	0.0263	0.0369	0.0348	0.0193	0.0171	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word ratio on the difference between LLM and word-based model performance with FF3 and mom6m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA10: Impact of negation word count with FF3 and mom12m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words count	0.0078***	0.0040***	0.0002	0.0106*	0.0119***	0.0080***	0.0043***	0.0146**
	(0.0008)	(0.0008)	(0.0007)	(0.0061)	(0.0007)	(0.0007)	(0.0005)	(0.0061)
beta	0.0026**	0.0028**	0.0006	0.0141	0.0020**	0.0022**	0.0000	0.0135
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0441***	-0.0432***	-0.0549***	0.2947***	-0.0005	0.0005	-0.0113***	0.3384***
	(0.0024)	(0.0024)	(0.0019)	(0.0181)	(0.0020)	(0.0020)	(0.0015)	(0.0183)
bm	0.0017	0.0018	0.0019**	-0.0069	-0.0000	0.0000	0.0002	-0.0086
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0087)
mom12m	-0.0076***	-0.0069***	-0.0053***	-0.0130**	-0.0048***	-0.0041***	-0.0025***	-0.0102*
Constant	0.0586***	0.0536***	0.0493***	-0.0463***	0.0263***	0.0212***	0.0169***	-0.0787***
	(0.0011)	(0.0010)	(0.0008)	(0.0079)	(0.0009)	(0.0009)	(0.0006)	(0.0080)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473	1,306,473
Adj R-squared	0.0316	0.0263	0.0369	0.0348	0.0193	0.0171	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word count on the difference between LLM and word-based model performance with FF3 and mom12m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA11: Impact of negation word ratio with FF3 and mom36m factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words ratio	0.0059***	0.0027***	-0.0006	0.0072	0.0102***	0.0070***	0.0037***	0.0115*
	(0.0008)	(0.0008)	(0.0006)	(0.0059)	(0.0007)	(0.0007)	(0.0005)	(0.0059)
beta	0.0022*	0.0027**	-0.0003	0.0218**	0.0023**	0.0028***	-0.0003	0.0218**
	(0.0012)	(0.0012)	(0.0010)	(0.0091)	(0.0010)	(0.0010)	(0.0007)	(0.0092)
mvel1	-0.0500***	-0.0480***	-0.0588***	0.2670***	-0.0030	-0.0010	-0.0118***	0.3140***
	(0.0025)	(0.0025)	(0.0020)	(0.0189)	(0.0021)	(0.0021)	(0.0016)	(0.0191)
bm	0.0022*	0.0021*	0.0021**	-0.0077	0.0008	0.0007	0.0006	-0.0091
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mom36m	0.0005	0.0004	-0.0007	0.0068	0.0006	0.0006	-0.0005	0.0069
Constant	0.0598***	0.0545***	0.0509***	-0.0466***	0.0258***	0.0205***	0.0168***	-0.0806***
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0086)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470	1,219,470
Adj R-squared	0.0323	0.0269	0.0381	0.0361	0.0202	0.0175	0.0125	0.0406

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word ratio on the difference between LLM and word-based model performance with FF3 and mom36m factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA12: Impact of negation word ratio with FF3 and industry factors

	SESTM				Word2Vec			
	OPT	RoBERTa	BERT	FinBERT	OPT	RoBERTa	BERT	FinBERT
neg words ratio	0.0069***	0.0034***	-0.0001	0.0109*	0.0113***	0.0078***	0.0042***	0.0153***
	(0.0008)	(0.0008)	(0.0006)	(0.0058)	(0.0007)	(0.0006)	(0.0005)	(0.0059)
beta	0.0032***	0.0033***	0.0011	0.0149*	0.0023**	0.0025**	0.0002	0.0140
	(0.0012)	(0.0012)	(0.0009)	(0.0088)	(0.0010)	(0.0010)	(0.0007)	(0.0089)
mvel1	-0.0513***	-0.0496***	-0.0601***	0.2840***	-0.0049**	-0.0033*	-0.0137***	0.3303***
	(0.0023)	(0.0023)	(0.0019)	(0.0175)	(0.0020)	(0.0019)	(0.0014)	(0.0176)
bm	0.0013	0.0013	0.0016*	-0.0073	-0.0003	-0.0002	-0.0000	-0.0088
	(0.0012)	(0.0011)	(0.0009)	(0.0086)	(0.0010)	(0.0010)	(0.0007)	(0.0087)
sic2	0.0002*	0.0003*	0.0004***	-0.0008	0.0000	0.0000	0.0001	-0.0010
Constant	0.0490***	0.0430***	0.0340***	-0.0038	0.0263***	0.0203***	0.0112***	-0.0265
	(0.0071)	(0.0071)	(0.0057)	(0.0532)	(0.0060)	(0.0059)	(0.0043)	(0.0537)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495	1,306,495
Adj R-squared	0.0315	0.0262	0.0369	0.0348	0.0193	0.0170	0.0117	0.0392

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of negation word ratio on the difference between LLM and word-based model performance with FF3 and industry factors where we use LLM signals minus word-based signals multiplied by next period return as the dependent variable. The first 4 columns show the results with SESTM as a word-based model benchmark, and the rest 4 columns show the results with Word2Vec as a word-based model benchmark

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Table IA13: Difference in OPTTK1024 and W2V

	FF3	FF3+Momentum				FF3+Industry
token type	0.0122***	0.0122***	0.0123***	0.0123***	0.0111***	0.0122***
	(0.0014)	(0.0014)	(0.0014)	(0.0014)	(0.0015)	(0.0014)
beta	0.0031***	0.0031***	0.0029***	0.0027***	0.0027***	0.0031***
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)
mvel1	-0.0064***	-0.0060***	-0.0031	-0.0016	-0.0052**	-0.0065***
	(0.0019)	(0.0019)	(0.0019)	(0.0020)	(0.0021)	(0.0019)
bm	-0.0003	-0.0003	0.0001	-0.0000	0.0006	-0.0003
	(0.0009)	(0.0009)	(0.0009)	(0.0009)	(0.0010)	(0.0009)
mom1m		-0.0013**				
		(0.0005)				
mom6m			-0.0050***			
			(0.0006)			
mom12m				-0.0054***		
				(0.0006)		
mom36m					0.0001	
					(0.0006)	
sic2						0.0001
						(0.0001)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,324,399	1,324,399	1,324,399	1,324,377	1,236,214	1,324,399
Adj R-squared	0.0182	0.0182	0.0182	0.0182	0.0192	0.0182

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of truncation in token length on the difference between OPTTK1024 model and Word2vec model performance. We use OPTTK1024 signals minus Word2vec model signals multiplied by next period return as the dependent variable. We include Fama-French 3 factors, momentum factors, and industry factors and other controls as independent variables. The sample period is from 2004 to 2019.

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Table IA14: Difference in OPT and W2V

	FF3	FF3+Momentum				FF3+Industry
token type	0.0118***	0.0118***	0.0118***	0.0118***	0.0107***	0.0118***
	(0.0015)	(0.0015)	(0.0015)	(0.0015)	(0.0015)	(0.0015)
beta	0.0030***	0.0031***	0.0028***	0.0027***	0.0028***	0.0031***
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)
mvel1	-0.0097***	-0.0093***	-0.0064***	-0.0050**	-0.0086***	-0.0097***
	(0.0019)	(0.0019)	(0.0020)	(0.0020)	(0.0021)	(0.0019)
bm	-0.0007	-0.0007	-0.0003	-0.0004	0.0003	-0.0007
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)
mom1m		-0.0011*				
		(0.0006)				
mom6m			-0.0051***			
			(0.0006)			
mom12m				-0.0053***		
				(0.0006)		
mom36m					0.0002	
					(0.0007)	
sic2						0.0000
						(0.0001)
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs	1,324,399	1,324,399	1,324,399	1,324,377	1,236,214	1,324,399
Adj R-squared	0.0178	0.0178	0.0179	0.0179	0.0191	0.0178

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents regression results examining the impact of truncation in token length on the difference between OPT model and Word2vec performance. We use OPT signals minus Word2vec signals multiplied by next period return as the dependent variable. We include Fama-French 3 factors, momentum factors, and industry factors and other controls as independent variables. The sample period is from 2004 to 2019.

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